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GEARS, BEVEL AND SPUR, REQUIREMENTS FOR



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DEPARTMENT OF DEFENSE
Washington, DC 20301

GEARS, BEVEL AND SPUR, REQUIREMENTS FOR

MIL-STD-1898(AT)

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FOREWORD

Procedures covered by this standard shall be used to provide acceptance criteria covering material and hardness requirements for ferrous-type gears, and inspection acceptance criteria for the manufacture of spur and bevel gears. Also covered are assembly procedures for bevel-type gears into a final mounting position. In addition, illustrations are provided depicting methods of measuring backlash, various tooth patterns obtainable, and corrective actions necessary to achieve proper tooth patterns and backlash during assembly operations.

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as required additional shooting gear or assemblies

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1. SCOPE

1.1 Purpose. The purpose of this standard is to provide criteria which shall be used as acceptance criteria for spur and bevel type gears, including procedures for adjusting bevel gears into final mounting positions at assembly.

1.2 Scope. This standard establishes the criteria for testing manufactured gears and proper mounting of bevel type gears. It also provides criteria for inspection, acceptance, and where applicable rework of defective gears.

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Standard. Unless otherwise specified, the following standard of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forms a part of this standard to the extent specified herein.

STANDARD
MILITARY

MIL-STD-1949 - Inspection, Magnetic Particle.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Spur gears.

3.1.1 Spur gear. A gear with teeth radially arrayed on the rim parallel to its axis. It is basically a cylinder with a number of regularly spaced mounds and depressions imposed upon its peripheral surface to form cogs, commonly called teeth.

3.1.2 Tooth profile. The tooth profile is the outside surface of the gear tooth. Basically spur gears will have tooth forms of the true involute type, however, in many applications this true involute profile must be slightly modified.

3.1.3 Tooth parallelism. Tooth parallelism is the parallelism of a gear tooth with respect to its rotational axis.

3.1.4 Tooth spacing. Tooth spacing is the distance between corresponding sides of adjacent teeth, measured along a line tangent to the base circle.

3.1.5 Tooth thickness. The gear tooth thickness is an arc distance on a circle which is concentric with the base circle.

3.1.6 Runout. Runout is the total difference of the off-center relation of the axis of the tooth profiles with respect to the rotational axis.

3.1.7 Tooth register. Tooth register of a gear is the dimension which specifies the relationship of gear teeth to each other on compound gears, or the relationship of the gear teeth to spline teeth. Tooth register dimensions are specified on applicable gear drawings.

3.1.8 Gear loaded face. The gear loaded face refers to the face of the gear tooth that makes contact with its mating gear tooth face. The unloaded gear face is the opposite side of the loaded or driving side of the tooth.

3.2 Bevel gears.

3.2.1 Bevel gear. A bevel gear is essentially a truncated cone with a number of cogs or teeth fastened on the cone surface. Two bevel gears, when rotated in respective non-parallel, intersecting axes should transmit uniform rotary motion. While in mesh, the gears roll on cones called pitch cones. The apex of this cone for each of the two mating gears, is at the point of intersection of the axes.

3.2.2 Reference mounting distance. This is the distance from the axis of one bevel gear to the thrust face of its mate as assembled in final mountings under no load. The final assembly is held to this dimension within the limits of accuracy imposed by shim lamination thickness. Original design is accomplished to insure that there will be sufficient shim stock for the extremes of tolerance build-up in the mountings.

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3.2.3 Operating distance. This is the actual distance, in final mountings from the axis of one bevel gear to the thrust face of its mate, for a specific operating condition. Three such operating distances may be specified on a bevel gear drawing. These are:

a. Normal operating distance.
b. Minimum operating distance.
c. Maximum operating distance.

The normal operating position will usually be that at which the greatest amount of running time will be accumulated in services. The minimum and maximum values are based upon the deflections inherent in the given assembly under the expected range of loads.

3.2.4 Theoretical tooth thickness. The thickness of tooth (measured at the large end) selected on the basis of a strength balance for the two mating gears of a set is the theoretical tooth thickness. These values are for zero backlash with the gears positioned at their respective reference mounting distances.

3.2.5 Basic value. The value to which a specified tolerance is applied is the basic value. For example, if a dimension is given as 1.490 + .XXX, -.XXX inches, the basic value is 1.490 inches.

3.2.6 Backlash. The amount by which the width of a tooth space exceeds the thickness of the engaging tooth, as measured by holding one gear fixed, rotating the mating gear, and reading the total movement on a dial indicator positioned normal to the active tooth surface.

3.2.7 Runout. Runout is the total variation in the distance between a surface of revolution and a reference surface, measured perpendicular to the surface of revolution. The surface of revolution of a bevel gear is the pitch cone.

3.2.8 Tooth spacing. The distance between corresponding sides of adjacent teeth taken in a plane which is perpendicular to the axis of the gear.

3.2.9 Pitch variation. Pitch variation of bevel and hypoid gears is the variation in spacing between corresponding sides of adjacent teeth, measurements preferably being taken near the middle of the tooth height at the midface.

3.2.10 Maximum index error. Maximum index errors of a gear is the greatest displacement of a tooth with respect to any other tooth measured near the pitch circle.

3.2.11 Required tooth contact. The required tooth contact determination is best accomplished by means of a testing machine. Proper contact involves three factors; namely, position, shape, and area of contact.

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3.2.12 Testing.

3.2.12.1 Testing. The inspection procedure involves a progressive proving process known in the industry as "testing". Testing is in addition to all basic inspections, such as for diameters, thicknesses, shoulders, bores, etc. Testing is a functional, rotational check and is introduced in the manufacturing process immediately following cutting the teeth in the green blank and continues through all subsequent operations which in any way affect the trueness, shape or size of the finished gear. "Testing" therefore is an essential phase of bevel gear inspection.

3.2.12.2 Inaccuracies. Smooth, quiet-running bevel gears must have correct tooth shapes, uniform spacing of teeth, and teeth concentric with the bore or shank. Inaccuracies in any of these conditions produces noise and rapid wear on high-speed or heavily-loaded installations, and non-uniform motion in the case of precision instruments and control equipment. Testing machines provide a practical method for determining that the gears are right before assembly, and for precisely controlling the quality of the gears during the stages of their manufacture.

3.2.12.3 Running test. In the testing machine, gears are mounted in the operating position and run under power and light brake load, simulating operation under normal running conditions in their final mountings. Smoothness and quietness of operation, tooth bearing, tooth size, surface finish, runout, and tooth spacing errors are all checked by the running test.

3.2.12.4 Tooth bearing. Tooth bearing is an indication of correct tooth shape both up and down and lengthwise on the tooth profile. It is that portion of the gear tooth surface which makes contact with its mate and can readily be observed by painting the teeth with a marking compound and running the gears for a few seconds in a bevel gear testing machine.

3.2.12.5 Bevel gear testing machines. Bevel gear testing machines are provided with all necessary adjustments to bring the axes of the gears into a relationship that optimum tooth bearing is obtained. By means of these adjustments, it is possible to determine how the gears will operate in and out of position in their final mountings and to measure directly the slight adjustments which must be made in the cutting or grinding machine to locate the tooth bearing correctly. To insure resetting of the testing machines to duplicate the original proven settings for a particular pair of bevel gears, special set-up gages are employed. The use of set-up gages is a practical solution to obtain, on repeat orders, the same quality as originally produced.

3.2.12.6 Bevel control gear. A bevel control gear is made to the necessary specification of size, mounting distance, tooth element accuracy and tooth bearing to serve as a working standard for the duplication of acceptable and interchangeable gears in production. Testing requires the use of the following types of master gears:

a. Master control gear. A bevel master gear used only for checking inspection control gears.

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b. Inspection control gear. A bevel master gear used only for checking finished production gears.

c. Green production control gear. A bevel master gear used only for checking gears in the green or unhardened state.

3.2.12.7 Trial or development pair. The routine for manufacture of bevel gears consists of first preparing a "trial or development pair" to be ultimately inspected and approved for operation in the final mounting. Following approval, several sets of master control gears are made which duplicate the "developed pair". These hardened control gears, tailored to satisfy requirements at various operations as production progresses, are then used in the testing machines to control quality at all stages of manufacture. Production gears are tested to the control pinion and the pinions tested to the control gears. This method insures positive control of tooth bearing and makes possible manufacture of quality bevel gears on a production basis.

3.2.13 Bevel gear measuring instruments.

3.2.13.1 Inspection devices. On certain high precision gear work, such as for instruments and for aircraft, extremely close tolerances of tooth spacing and concentricity must be held. Checking of these values by sight and sound on a testing machine is not sufficient; they must also be inspected by direct measurement. Individual elemental inspections must be made to prove setups, cutter action, machine performance, tooling, etc. For these purposes three inspection devices are employed; the composite error tester, the tooth spacing tester, and the concentricity tester.

3.2.13.2 Composite error test. The composite error test is a gear rolling operation for determining size variations, runout, trueness errors, tooth spacing errors, tooth bearing, and backlash.

3.2.13.3 Trueness errors. Trueness errors are represented as slight oscillations in the indicator as each tooth goes in and out of mesh during rotation in close engagement without load. These oscillations may or may not be uniform in magnitude and are caused by the modification in the tooth profile shape (necessary to localize the tooth bearings) and by tooth spacing errors. The oscillation caused by modification in the tooth profile shape is referred to as the profile wave. This will be approximately equal to the average value of all the individual oscillations around the gear.

3.2.13.4 Spacing errors or tooth defects. Thick or thin teeth or the presence of dirt or chips are indicated by sudden jumps of the indicator either up or down, on top of, or in addition to the profile wave.

3.2.13.5 Runout. Runout is determined by rolling the gear with a mating member or control gear known to have no appreciable runout. Runout shows up as a gradual increase in the indicator reading up to a high point and a gradual decrease in the indicator reading to a low point, which is generally 180° around the circumference of the member being tested from the high point.

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Since the amount of the profile wave is the same for every tooth, it has nothing to do with runout. Therefore, it must be subtracted from the difference between the high and low reading to obtain the actual runout. Faulty arbors or improper set-up may show up as a runout in otherwise good gears.

3.2.13.6 Tooth spacing tester. This instrument measures the variation in spacing from tooth to tooth. Two contact points are employed, one a fixed stop, the other movable to actuate a dial indicator. In use, each tooth of the gear is brought to bear against the fixed stop with the movable point contacting the corresponding flank of the neighboring tooth. The dial indicator reading is noted and the operation is then repeated for all the teeth in the gear.

3.2.13.7 Concentricity tester. Concentricity of the pitch circle with the center about which the tester turns is measured by the concentricity tester. The runout is twice the eccentricity. This instrument employs two contact points 180° apart, one a fixed stop, the other actuating a dial indicator. The gear to be checked is mounted on a dead center held in the vise, and indicator readings are taken on diametrically opposite teeth with the contacts acting on corresponding sides of these teeth. The contact points are lifted while the gear is turned from tooth to tooth. The runout is equal to one-half the full indicator movement (FIM), which make this method of inspection extremely accurate.

NOTE: Because of the great amount of time required in inspecting gears by these last two devices, they are used primarily for periodic and not general production testing.

3.2.13.8 Special checking fixtures. Testing machines are preferred over special fixtures because they show how the gears will operate in service. They provide adjustments necessary for determining slight changes in set-up of the cutting machine for control of tooth bearing.

3.2.14 Checking of tooth profiles.

3.2.14.1 Tooth profiles. The checking of the tooth profiles is not carried out on bevel gears for two principal reasons. First, the shape of a bevel gear tooth varies at every position from end to end, instead of being constant as in a spur gear. Hence, a different reading is obtained at every checking position along the gear tooth and even a slight variation in checking position from gear to gear will give incorrect results. Second, even if this check were practical for bevel gears, it would not show the true contact along the entire length of the tooth, as shown by a running test in a testing machine.

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4. GENERAL REQUIREMENTS

4.1 General. This standard describes a series of methods, standards, and tasks to assure that, spur and bevel type gears meet a quality standard that specifies the required level of performance for tank-automotive material.

4.2 Objective. The objective as applied in this standard is to assure that spur and bevel gears are designed, manufactured and tested for required performance in a military environment. This standard has four methods of acceptance defined as follows, metallurgical examination, spur gear inspection requirements, bevel gear inspection and testing and assembly of bevel gears into their final mounting position.

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5. DETAILED REQUIREMENTS

5.1 Metallurgical standards.

5.1.1 Applicability. This section provides guidance in establishing metallurgical requirements relative to material inclusions and inspection criteria for surface (case) hardening of ferrous gears.

5.1.1.1 Classification. Gear classification shall conform to those designated on engineering drawings. Established levels of inspection shall be in accordance with the following categories of gear areas:

a. Class 1 gears. These are gears for heavy duty service which could approach and/or operate under loads reaching the physical limits or capability of the material. They must have the ability to withstand high bending and compressive stresses, shock or impact loads and resist wear.

b. Class 2 gears. These gears are for medium duty service which could operate for a major portion of the life under loads substantially below the physical properties of the material and must have the ability to withstand moderate bending and compressive stresses, moderate shock or impact loads and resist wear.

c. Class 3 gears. The severity of service that these gears are subjected to is considered to be light as compared to Class 1 and 2 gears. They are relatively lightly loaded and used in application where resistance to wear is the primary factor in obtaining adequate gear life.

5.1.1.2 Gear areas. Figures 1a and 1b depict specific areas of gears and are defined as area A, B and C relative to the compressive stresses, bending, shock or impact loads anticipated.

a. Class 1. Areas shall be in accordance with figure 1a and 1b.

b. Class 2. Gear teeth and splines shall be areas A or B. The remainder of the gear shall be considered area C.

c. Class 3. All areas shall be considered gear area C.

5.1.1.3 Acceptance criteria. The criteria to be used for determining the acceptability of gears which have imperfections are specified in tables I, II and III, and as specified below.

a. Acceptable indications: Magnetic particle indications associated with nonmetallic inclusions which do not exceed the following:

1. Nonrateable indications: Any one indication, 0.06 inch or less in size which does not occur in axial strings.

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2. Rateable indications: Indications which conform to the requirements of the applicable tables for the following gears:

   a. Class 1 - Table I.

   b. Class 2 - Table II.

   c. Class 3 - Table III.

b. Unacceptable indications are specified as follows:

   1. Cracks, laps, bursts, etc., and grinding checks.

   2. Through wall indications.

   3. Indications breaking over a machined edge on Class I and II gears.

5.1.1.4 Hardness. Gear hardness shall be as specified on the engineering drawing. The component drawing or the procurement document shall also specify the following:

a. Method of magnetization including magnetizing current, type, magnitude and direction.

b. Method of particle application.

c. Type of particle (dry or wet visible or wet flourescent).

d. Concentration of suspension.

5.1.2 Test methods.

5.1.2.1 Test method. Visual and magnetic particle inspection shall be performed to determine conformance to the acceptance criteria specified in 5.1.1.3 for the class of gear specified. Magnetic particle inspection shall be conducted in accordance with MIL-STD-1949.

5.1.2.2 Hardness testing. Upon the completion of all heat treat and case hardening operations, gear class specifications shall conform to the following criteria and shall be measured as shown on figure 2.

a. Class 1:

   1. Tooth surface hardness measurements shall be taken on a minimum of 3 gear teeth located approximately 120° apart on the gear tooth area.

   2. Tooth surface hardness measurements shall be taken with the Rockwell 15 N scale or equally sensitive measuring device, and suitable fixtures, on the tooth profile of finished machined gears.

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3. Case hardness measurements on carburized or nitrided gears or case hardness measurements on the tooth outside diameter shall be taken using the Rockwell 30N scale, or equally sensitive measuring device, and suitable fixture.

b. Class 2 and Class 3:

1. Tooth surface hardness measurements shall be performed on the load side of each tooth face at the pitch diameter using the Rockwell 15N scale or equally sensitive measuring device, unless otherwise specified on engineering drawings.

2. Spiral and bevel gears shall conform to case hardness requirements specified on the engineering drawings.

c. Core hardness measurements on all gear classes shall be taken on a sectioned gear tooth or a sectioned test specimen tooth.

d. Case depth, "effective" case depth on all gear classes shall be measured as shown in figure 2. Measurements shall be made on a sectioned gear tooth or a sectioned test specimen.

5.1.2.3 Rework criteria. Gear areas containing indications may be reworked by blending, provided the minimum dimensions and surface finish requirements of the engineering drawings are met. Except for the specific methods of rework stated herein, all other methods used for general rework shall require approval of the Material Review Board, prior to use.

5.1.2.4 Blending. Indications shall be smoothly blended into the surrounding area by means of a carbide burring bit or equivalent tool. All blending shall conform to the limits specified in table I, and figures 1a and 1b.

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TABLE I. Acceptance limits for class 1 gears.

Gear Area A Any indication (rateable or unrateable) shall subject the gear to rejection.
Gear Area B 1. Surface indications 0.25 inch or less and sub-surface indications 0.37 inch or less shall be acceptable. 2. If the distance of closest approach between indications in an axial direction is less than the length of the larger indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication. 3. Parallel surface or sub-surface indications shall be separated by 0.25 inch minimum. 4. Nor more than 2 acceptable indications shall be permitted in a 1.50 inch diameter circle. 5. Not more than 4 indications shall be permitted per gear in the B areas.
Gear Area C 1. Indications 0.50 inch or less shall be acceptable. 2. If the distance of closest approach between indications in an axial direction is less than the length of the larger indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication. 3. Parallel indications shall be separated by 0.25 inch minimum. 4. Not more than 4 indications shall be permitted per part in the C areas.

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TABLE II. Acceptance limits for Class 2 gears.

Gear Areas A and B The presence of any rateable indications shall subject the gear to rejection.
Gear Area C 1. Surface indications 0.25 inch or less and sub-surface indications 0.37 inch or less shall be acceptable. 2. If the distance of closest approach between indications in an axial direction is less than the length of the larger indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication. 3. Parallel surface or sub-surface indications shall be separated by 0.25 inch minimum. 4. Not more than 2 acceptable indications shall be permitted in a 1.50 inch diameter circle. 5. Not more than 6 acceptable indications shall be permitted per gear in the C areas.

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TABLE III. Acceptance limits for Class 3 gears.

1. Indication length:

   a. Indications 0.75 inch or less shall be acceptable.

   b. In fillet radii with radii 0.25 inch or less, indications 0.06 inch or less shall be acceptable.

   c. The total length of all indications in any one square inch shall not exceed one linear inch.

2. If the distance of closest approach between indications in an axial direction is less than the length of the smaller indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication.

3. Parallel surface indications shall be separated by 0.18 inch minimum. Parallel sub-surface indications shall be separated by 0.12 inch minimum.

4. Indications shall not extend over a machined edge more than 0.030 inch or 15 percent of wall thickness, whichever is less.

5. Gear teeth and splines:

   a. Indications shall not be acceptable in the gear or spline root or between root and tooth (spline) pitch diameter.

   b. Indications extending over the edges of splines or gear tooth profiles 0.015 inch or less shall be acceptable.

   c. One hairline surface indication per tooth between pitch diameter and tip of spline or gear tooth shall be acceptable provided the length of the indication does not exceed 15 percent of tooth length up to 0.060 inch maximum.

   d. Sub-surface indications per tooth between pitch diameter and tip of spline or gear tooth shall be acceptable provided the length of the indication does not exceed 20 percent up to 0.090 inch maximum. Not more than 15 percent of the splines or gear teeth shall contain such indications.

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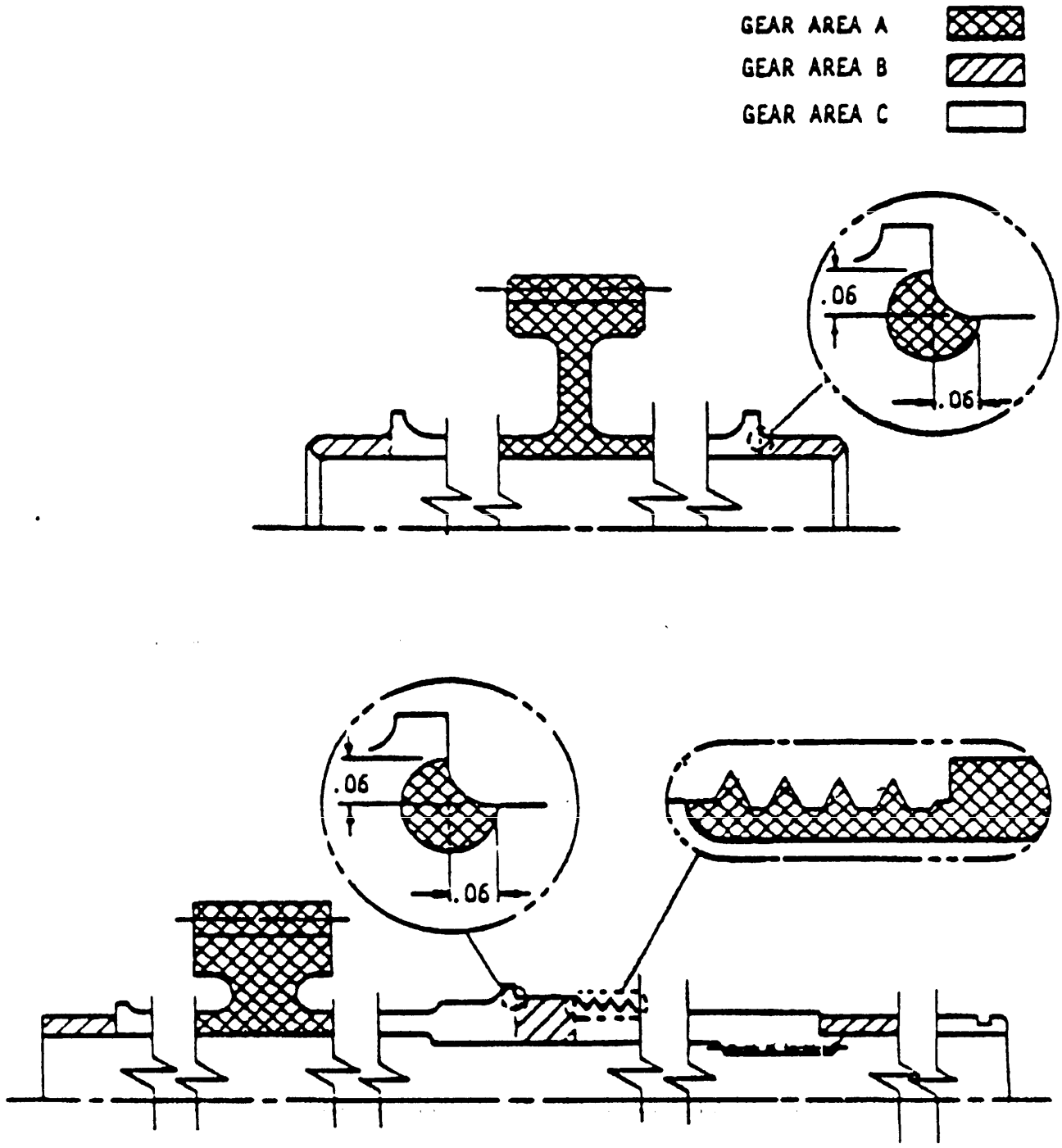


FIGURE 1a. Typical class 1 gear areas.

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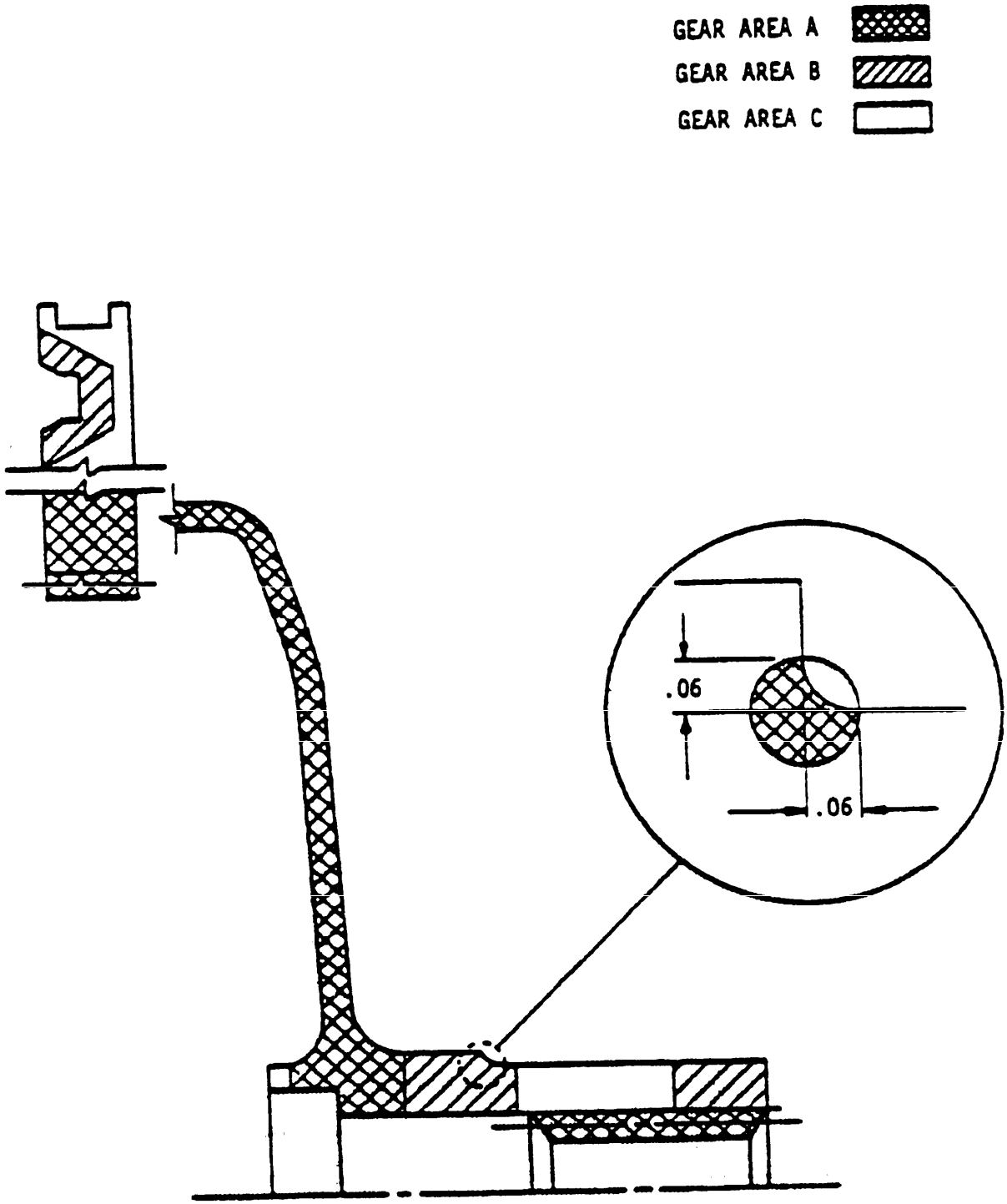
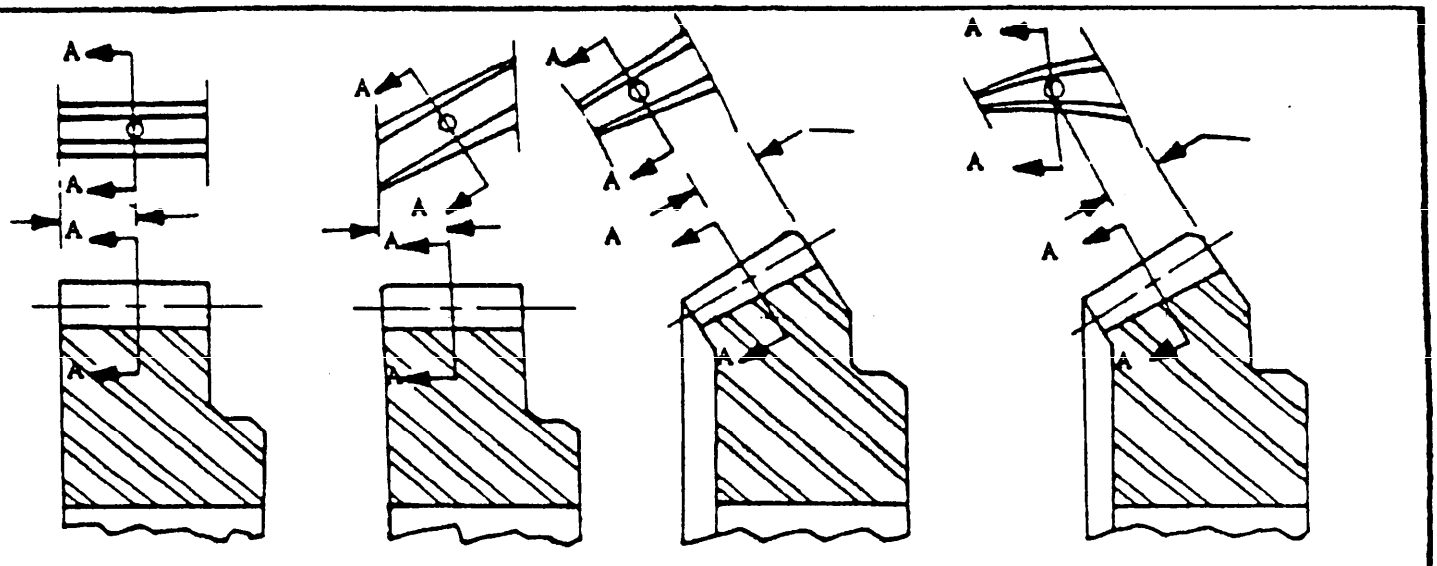
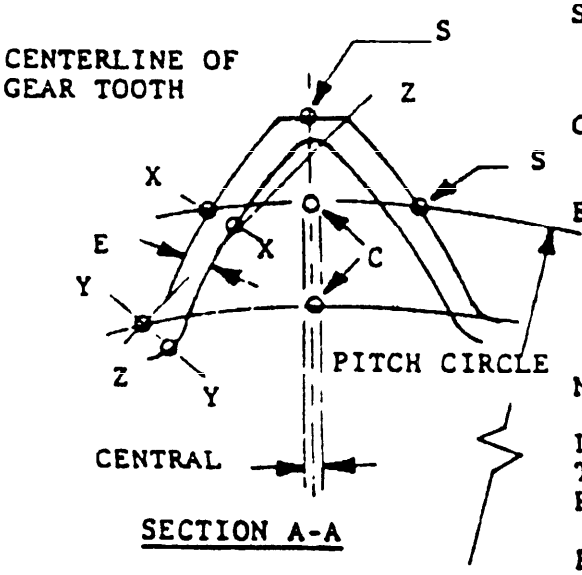


FIGURE 1b. Typical class 1 gear areas.

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( ) SPUR GEAR ( ) HELICAL GEAR CUT NORMAL TO HELIX ANGLE ( ) STRAIGHT BEVEL GEAR ( ) SPIRAL BEVEL GEAR CUT NORMAL TO SPIRAL ANGLE ( ) HYPOID GEAR CUT NORMAL TO SPIRAL ANGLE



SECTION A-A

S = SURFACE HARDNESS _____
( ) TOP LAND OF TOOTH
( ) NORMAL TO TOOTH SURFACE AT PITCH CIRCLE

C = CORE HARDNESS _____
( ) AT PITCH CIRCLE
( ) AT ROOT CIRCLE

E = EFFECTIVE CASE DEPTH _____
CASE HARDNESS _____
( ) TRAVERSE ALONG LINE X-X NORMAL TO TOOTH SURFACE
( ) TRAVERSE ALONG LINE Y-Y PERPENDICULAR TO LINE Z-Z

NOTES:
1. EFFECTIVE CASE DEPTH IS THE LENGTH OF LINE X-X OR LINE Y-Y MEASURED FROM THE SURFACE TO THE LAST POINT TOWARD THE CORE HAVING THE REQUIRED MINIMUM HARDNESS OF ROCKWELL C ____.
2. LINE Z-Z IS TANGENT TO ROOT FILLET AND PASSING THROUGH OPPOSITE TIP CORNER.

LEGEND: (X) APPLICABLE (-) NOT APPLICABLE

FIGURE 2. Hardness determination methods.

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5.2 Spur gears.

5.2.1 Applicability. This section prescribes procedures which will accurately measure the dimensional characteristics of involute tooth from spur gears including runout, parallelism, and composite tooth errors to assure compliance to the applicable part drawings.

5.2.1.1 Spur gear errors. Involute type spur gear errors occur in the shape of the teeth, and shape of the base circle, relationship or position of the teeth on the base circle, and the relationship of the base circle to the rotational center of the gear. The fundamental errors in gears covered in this standard are:

a. Tooth profile error.
b. Tooth parallelism error.
c. Tooth spacing error.
d. Tooth thickness error.
e. Runout and tooth action error.
f. Tooth register.

5.2.1.2 Tooth profile error. Basically, spur and helical gears will have tooth forms of the true involute type, but in many applications this true involute profile must be slightly modified. The tolerance band(s) into which the actual measured tooth profile(s) of the loaded face(s) must fit shall be as specified on the gear drawing. Unacceptable tooth profile error of a gear tooth is the amount the actual measured tooth profile is outside the specified tolerance band. The actual profile check must start on the specified true involute form (TIF) diameter at any point within the tolerance band. As an example, in the acceptable profile checks shown in the upper diagram of figure 3, the actual check was started on the minus side on the parallel band and on the plus side in the modified band. Both are acceptable because both fit entirely within the tolerance band. In addition to the amount of profile deviation, the rate of deviation (or slope) is also defined on the gear drawing by a note which reads: "Rate of departure from the measured mean profile must not exceed X.XXX inches in 25% of the total roll from the true involute diameter to the outside diameter." It is possible for the actual measured tooth profile to be completely within the tolerance band but exceed the maximum acceptable rate of departure (slope). This condition is shown in the lower (unacceptable) diagram of figure 3.

5.2.1.3 Tooth parallelism error. Tooth parallelism error of a gear tooth with respect to its rotational axis may be described as the deviation of an element of the tooth surface from a straight line toward a concave or convex curve. The tooth parallelism is controlled by a note on the gear drawing that reads: "Loaded tooth surface parallelism with axis within X.XXXX. Maximum crown permitted X.XXXX." It shall be understood that the parallelism requirement refers to elements of the tooth surface at a constant distance from the rotational axis (see figure 4). The acceptable crown should be a uniform rise and fall of the tooth surface in addition to the acceptable out of parallelism. The rise maximum should be within the center 1/3 of the face width.

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Method of Code No. Categories Defects

5.2.1.4 Tooth spacing error. The tooth spacing error of a gear is the variation in the distance between corresponding sides of adjacent teeth, measured along a line tangent to the base circle with the point of tangency not between the two involute surfaces being measured (see figure 5). By definition, this distance is termed base pitch. It should be noted that the base pitch is constant anywhere along true involute curves, and is also equal to the length measured in an arc along the base circle between these same involute curves. Its numerical value is thus the circumference of the base circle divided by the number of teeth. The base pitch is given on the gear drawing. The tooth-to-tooth spacing error is defined as the maximum difference in base pitch between any two successive measurements.

5.2.1.5 Tooth thickness error. The gear tooth thickness is measured as an arc distance on a circle which is concentric with the base circle. The arc tooth thickness controls the backlash in two mating gears. The tooth thickness is most easily measured by measuring across two pins placed in opposite spaces on the gear. This method is actually measuring the chordal thickness.

5.2.1.6 Runout. Runout of a gear is the total difference between the high and low readings of a dial or recording indicator suitably arranged to denote the off-center relation of the axis of the tooth profiles with respect to the rotational axis. In the case of a gear whose base circle is a true circle but eccentric with respect to its rotational axis, runout is twice the eccentricity. Runout also includes the effect of non-circularity of the base circle such as scallops, oval shapes, etc.

5.2.1.7 Tooth action error. Tooth action error of a gear is the combination of profile error, tooth to tooth spacing error, accumulated spacing error, interference, etc. All these errors must be absorbed within the tooth action error as specified on the drawing.

5.2.1.8 Tooth register. Tooth register of a gear is the dimension which specifies the relationship of teeth on compound gears or the relationship of gear teeth to the spline teeth. In the case of planetary or epicyclic gear trains, the use of compound pinions (or any gear cluster having two or more sets of teeth on a common axis) requires close control over the relationship of the two sets of teeth. This is necessary to insure uniform load distribution among the pinions of such systems. Tooth register limits shall be as specified on drawings and illustrated in figure 17.

5.2.1.9 Hardening. Gears that have been accepted as meeting the requirements of this standard shall be hardened as specified on the applicable gear drawings, and shall meet the requirements of 5.1.

5.2.1.10 Finish. The surface finish of the gears shall be as specified on the applicable gear drawings. Gears shall be deburred and sharp edges removed.

5.2.1.11 Marking. Each gear shall have identification markings etched or stamped as specified on applicable part drawings.

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5.2.2 Test methods.

5.2.2.1 Classification. All gear error classifications shall conform to designated gear drawings. Inspection shall be in accordance with the following categories of involute type gear errors.

a. Tooth profile error
b. Tooth parallelism error
c. Tooth spacing error
d. Tooth thickness error
e. Runout and tooth action error
f. Tooth register

5.2.2.2 Acceptance criteria. Acceptance criteria for involute type spur gears shall be in accordance with the gear drawing. Fundamental involute type spur gear measurement errors shall occur within the allowable limits specified in the following paragraphs and figures 3 through figure 18.

5.2.2.3 Tooth profile error. To comply with 5.2.1.2 the tooth form of a gear shall be determined by traversing the tooth from its true involute form diameter to its outside diameter (or inside diameter in the case of internal gears) with a follower suitably arranged to describe a true involute curve with respect to the base circle of the gear being inspected. This follower shall be so mounted that any motion induced in the follower by a deviation in the tooth profile from a true involute form will be shown on a recording indicator.

a. Figure 6 shows a typical tooth profile chart as obtained on a "Fellows" involute gear checker. The straight lines represent a true involute form. On this chart, the distance between the fine lines represents .0002 inches. The arcs represent even angular increments of rotation or roll of the gear. The distance between the arcs represents three (3) degrees of roll. This roll angle is also referred to as the generating angle.
b. Referring to figure 7, it may be seen that the roll angle or generating angle depends only upon the diameter of the base circle, and the radius in question. Since the follower is spring loaded against the tooth surface, it moves in the direction of tooth material when it leaves the tip of the gear. If we call the start of the involute check (point A on figure 6) the start of a true involute, the chart (figure 6) shows that the tooth deviates from a true involute by having excess material at the tip. A cross section view of such a tooth is shown in figure 8. There are two types of profile limit bands.

    1. The parallel band (see figure 9a).
    2. The modified profile band (see figure 9b).

c. In order for a tooth profile to be considered as satisfactory, its chart must meet two general requirements:

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1. All tooth profile lines must be within the area defined by the band shown on the involute diagram.

2. All tooth profile lines must be smooth and uniform. Concave, wavy or irregular tooth profile lines are not acceptable when the cavity, wave, or irregularity represents a rate of departure from the measured mean profile line in excess of the rate of departure specified on the gear drawing (figure 3 which illustates both acceptable and unacceptable profile charts).

d. In matching the actual measured involute profile to the tolerance band, the start of the actual measured involute profile diagram may be placed at any point on the True Involute Form (TIF) diameter as represented on the chart within the confines of the band at that diameter.

5.2.2.4 Tooth parallelism error. To comply with 5.2.1.3 the tooth parallelism and crown limits specified on the gear drawing apply only to the gear loaded face(s). On gear drawings where the acceptable crown is not given, it should be taken as one half (1/2) the acceptable parallelism error. All tooth surfaces, loaded or unloaded, not otherwise controlled on the gear drawing shall be parallel with the axis within 0.002 inches per inch. In making the parallelism check, the gear must be supported in relation to the axis. The indicator must travel in a line parallel to the axis and contact the tooth surface in the area of the pitch point (rolling pitch diameter). Figure 3 illustrates this technique. Because of machining operations performed on gears after the teeth have been finished, it is possible for a finished gear to have an axis of rotation that is not parallel to the axis of the base cylinder of the teeth (see figure 10).

a. If the parallelism of the surfaces of each tooth is checked with respect to the rotational axis of the gear, and the measurements plotted along a line representing the teeth, the results will approximate a sine wave as shown in figure 11.
b. Figure 11 illustrates the parallelism check of a 15 tooth gear. It can be readily seen that some of the gear teeth are within the parallelism tolerance. However, if any three equally spaced gear teeth are examined at least one of the teeth is in fact out of tolerance. For example look at teeth numbers 1, 6, and 11. Teeth number 1 and 6 are almost within tolerance while tooth number 11 is 0.001 inches over the tolerance. Therefore when checking parallelism, check 3 gear teeth approximately 120° apart. Only straight or crowned gear tooth surfaces are acceptable, with no concavity allowed. Figures 12a and 12b illustrate acceptable and unacceptable tooth surfaces.

5.2.2.5 Tooth spacing error. To check for tooth spacing errors of 5.2.1.4, tooth to tooth error readings may be taken by using a device consisting of a pair of fingers, one fixed and the other actuating a dial indicator. Both are mounted on a movable block or slide with means to accurately return the slide to the same position. The position of the slide, when the readings are being taken, should be such that the fingers contact

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the teeth on a line tangent to the base circle. A spring tension device is desirable to keep a uniform pressure of the gear against the fixed extender (see figure 5). The indicator is set at zero for the first pair of teeth and all readings between successive pairs of teeth are relative to the first reading. Table IV illustrates the method of showing readings recorded for a twelve tooth gear.

NOTE: The difference between successive readings denotes spacing error. The reliability of this check is limited to the sensitivity of the equipment and skill of the operator.

5.2.2.6 Tooth thickness error (normally measured over two pins). To check tooth spacing error of 5.2.1.5, the pin size to be used in making measurements over, or between, two pins should be listed on the gear drawing. Only standard gear measuring pins should be used. They are correct at 68°F. The measurements between, or over, two pins must be within the tolerance specified on the gear drawing over the full circumference of the gear. For gears with even numbers of teeth, the pins should be placed in diametrically opposite tooth spaces. For gears with odd numbers of teeth, the second pin should be placed in the space either side of the tooth that is diametrically opposite the first pin. The measurements that are listed on the gear drawing take into account the odd or even numbers of teeth; no adjustment in the values is required.

5.2.2.7 Runout. Runout of a gear may be measured by observing, or graphically recording, the center-distance variation when rotated about its rotational axis tightly in mesh with a master gear, excluding the effect of any center-distance variation caused by profile errors, interference, etc. Figures 13a, 13b, and 13c illustrate runout caused by eccentricity, out of round, and scallops respectively (see 5.2.1.6).

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TABLE IV. Measurement readings for tooth spacing errors.

Column 1 Tooth number	Readings in ten-thousandths of an inch Column 2 Indicator reading	Column 3 Difference between successive indicator readings of column 2
1	0	
2	+2	+2
3	+3	+1
4	+4	+1
5	+2	-2
6	0	-2
7	0	0
8	+1	+1
9	-2	-3
10	-1	+1
11	-2	-1
12	-1	+1
1	0	+1
		SUM 0 (see note below)

NOTE: Any tooth spacing which are too large must be offset by others which are too small. The sum of all the differences (column 3) must be zero. If errors are made in reading the indicator, the sum of column 3 will not equal zero. If the sum of column 3 deviates from zero by more than 0.0002 inch, the check must be repeated. The largest number (regardless of sign) in column 3 is the maximum tooth to tooth spacing error. This value shall be equal to or less than the value specified on the gear drawing.

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5.2.2.8 Tooth action error. The tooth action error specified in 5.2.1.7 may be measured by observing, or graphically recording the center-distance variation when rotated on its rotational axis tightly in mesh with a master gear, excluding the effects of any center-distance variation caused by runout. Figures 14a and 14b illustrate tooth action error. Errors in runout, spacing, and profile each cause a characteristic deviation from a straight line on a graphical recording of the center-distance variation when rotated tightly in mesh with a master gear.

a. Figures 13a, 13b, 13c, 14a, and 14b, show characteristic recording for each of these types of errors. Figures 13a, 13b, and 13c show typical cases of pure runout. Figures 14a and 14b show typical cases of pure tooth action errors. An actual gear, when tested by this method, will show up a combination of the errors. Figure 15 shows a typical graph of a gear having combined errors. The maximum runout and the maximum tooth action error should be inspected according to this figure to see that the values of each one of these errors is not greater than the limits shown on the gear drawing. It is to be noted that the runout (as shown in figure 15) is the maximum variation of the imaginary center line of the tooth action error band. The particular pattern shown on the tooth action diagram of the gear drawing is included only to illustrate the specified limits. Figure 16 shows a case of an abrupt change in the tooth action pattern which is an unusual condition and is a special case of runout. Such a condition can arise as a result of improper cutting, heat treatment distortions, or because of improper grinding. In order to define the permissible extent of such abrupt changes within a very few teeth, the following rule shall apply: Abrupt changes in the tooth action pattern, where the rate of change exceeds 50% of the runout limits within 10% of the number of teeth in the gear, are not acceptable (see figure 16). When the graphical recording of a gear test shows that the tooth action error limit specified on the drawing has been exceeded, the gear in question shall then be checked separately for:

    1. Tooth profile error.
    2. Tooth to tooth spacing error.
    3. Measurement over, or between, two pins.

b. These supplementary checks should be made in the area having unsatisfactory tooth action. Attention should be paid to such items as burrs and dirt, since they can result in unsatisfactory tooth error. If these supplementary checks indicate proper tooth profile, spacing, and measurement over or between pins, the gear is acceptable. If not, the gear is unsatisfactory.

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5.2.2.9 Tooth register. To measure tooth register specified in 5.2.1.8, (shown in figure 17) measurements of tooth register shall be taken at the diameters specified because a small error in the diameters will produce an appreciable error in the register measurement due to the tooth curvature. It should be noted that the tooth register dimension lines are parallel to the tooth centerline of the smaller gear and are not radial lines. Therefore, the method employed to measure tooth register error shall be one that actually measures the perpendicular distance these lines.

a. In the usual case where one gear of the cluster has a larger pitch diameter, the tooth register limits shall be checked as close as possible to the X.XXXX diameter of that gear. Fixtures employed to check tooth register shall incorporate the method shown in figure 18.

b. All fixtures used to check tooth register, shall provide a fixed stop for contacting the loaded face of the tooth with the gage point on gear A as shown. The contacting surface of the fixed stop must be tangent to the tooth profile of gear A (the smaller gear) at its X.XXXX diameter and parallel to the centerline of both gears.

c. When a mechanical gaging or measuring device is used to check the tooth register limits, the checking point shall be on the loaded face of a tooth profile on gear B (the larger gear) as close as possible to the X.XXXX diameter on the engineering drawing. The tangent to this check point must be parallel to the centerline of both gears.

d. Dimensions X_1 and X_2 shall be measured perpendicular to the centerline of both gears.

e. When an optical gaging method is used, the tooth register limits specified on the engineering drawing can be checked or measured directly.

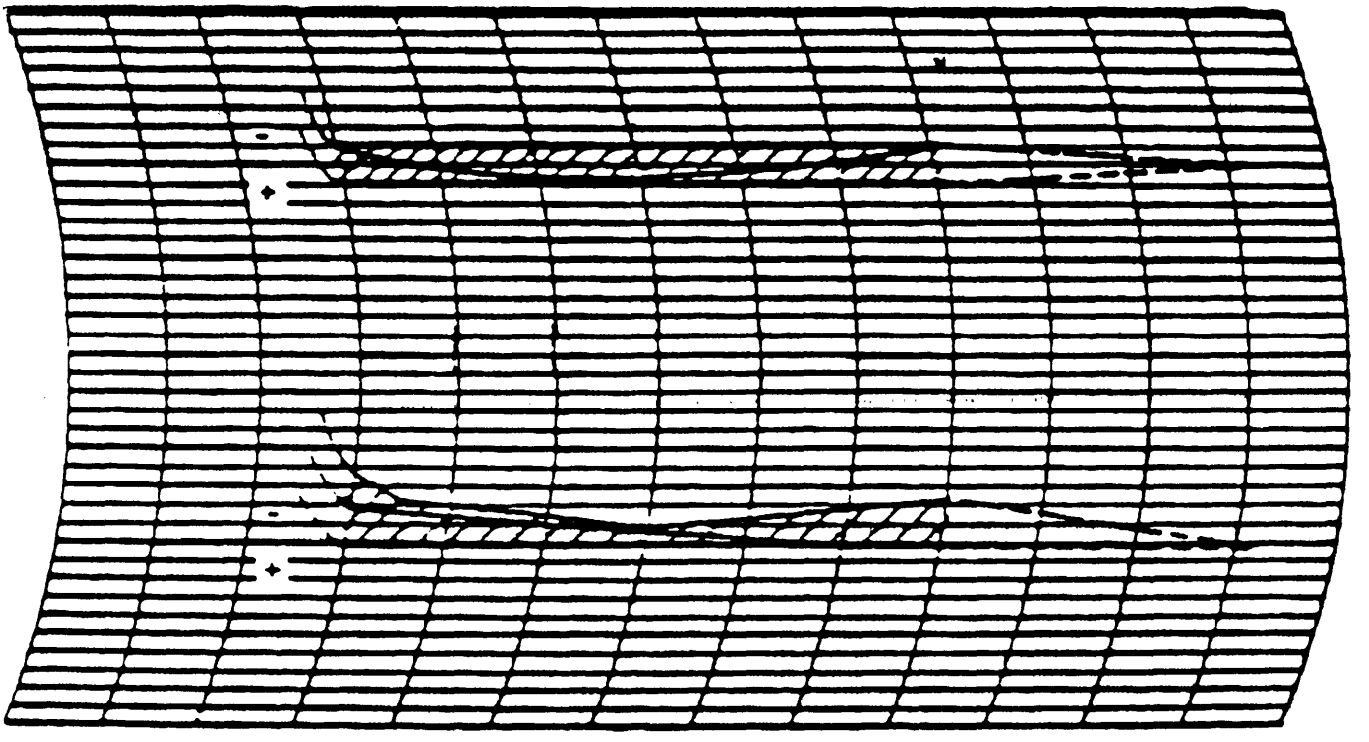
5.2.2.10 Hardening. Spur gears shall be heat treated and hardened as specified on the applicable gear drawings. If the gears are hardened all over, hardness readings shall be taken at various points on the side of the gear. If only the tooth surface requires hardening, the hardness checks shall be made as follows. A minimum of three (3) samples shall be selected from each heat treat lot for determination of effective case depth, core hardness and surface hardness. Samples shall be processed by the same heat treat process procedure used for production parts. Inspection and hardness tests shall be performed in the areas shown in 5.1.

5.2.2.11 Finish. Surface finishes specified on the applicable part drawings shall conform to the specified requirements. Burrs or sharp edges on any part of the gears are unacceptable.

5.2.2.12 Marking. Gears shall be permanently marked as specified on the applicable part drawings.

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ACCEPTABLE



UNACCEPTABLE

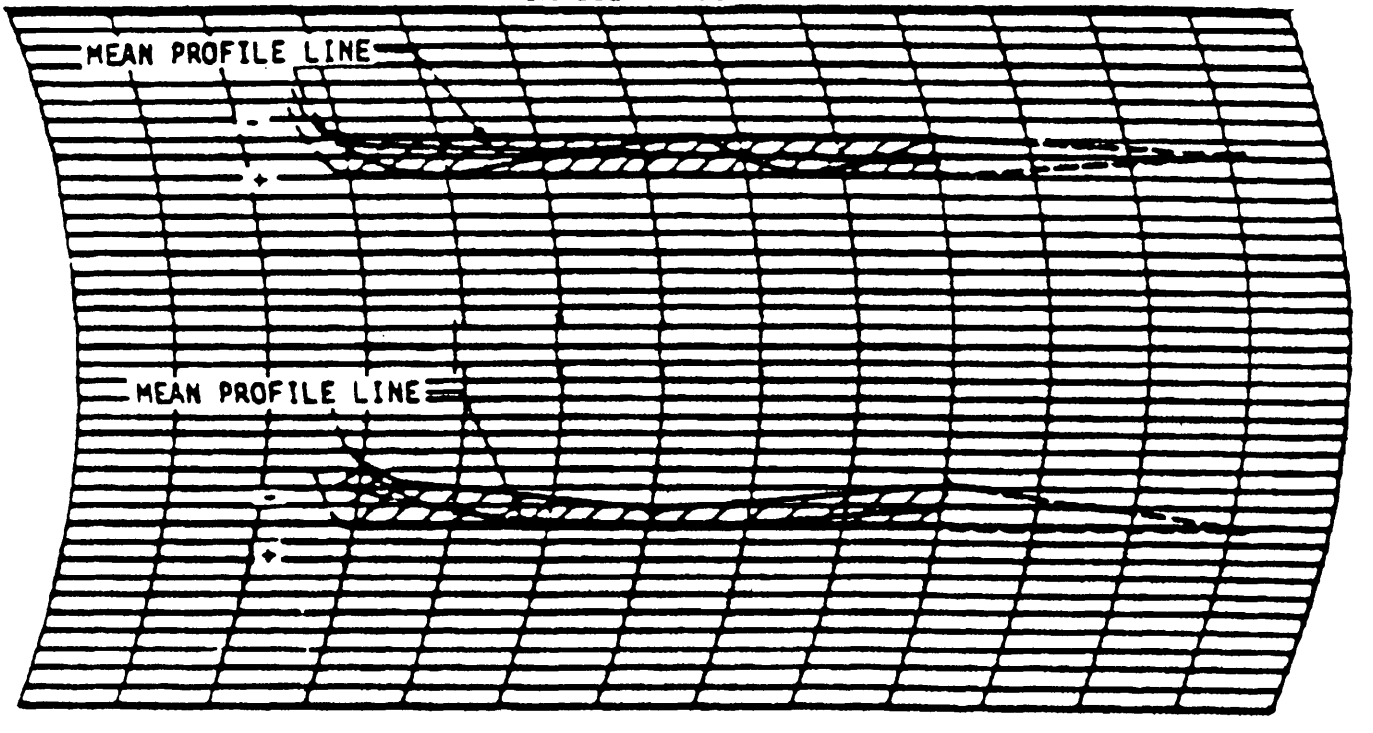


FIGURE 3. Tooth profile check.

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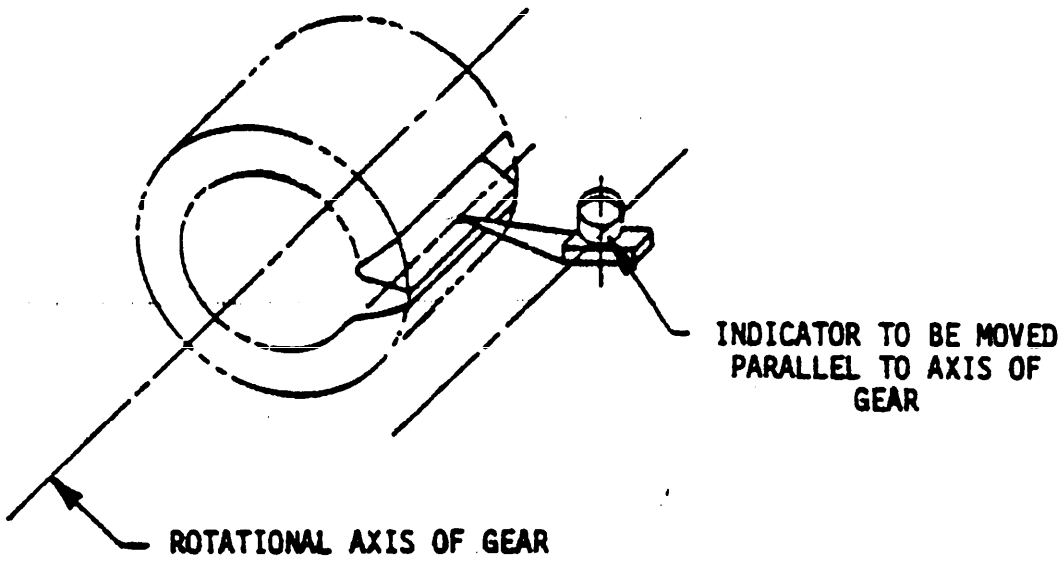


FIGURE 4. Tooth parallelism measurement.

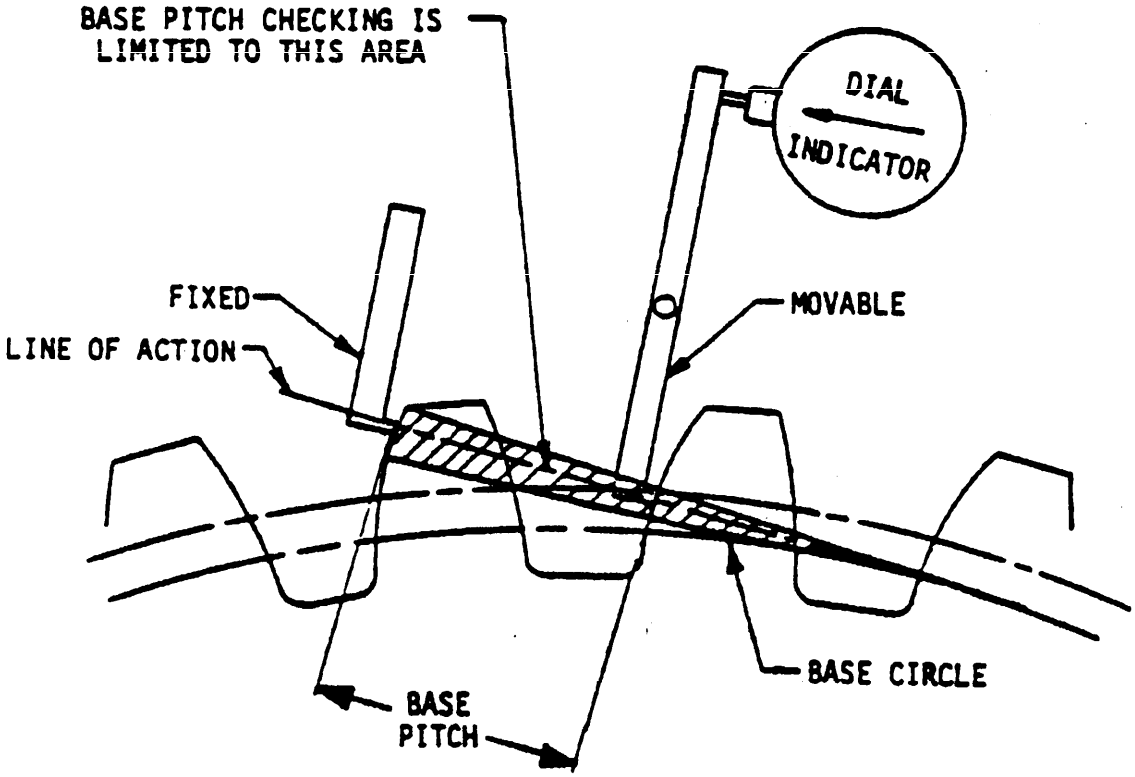
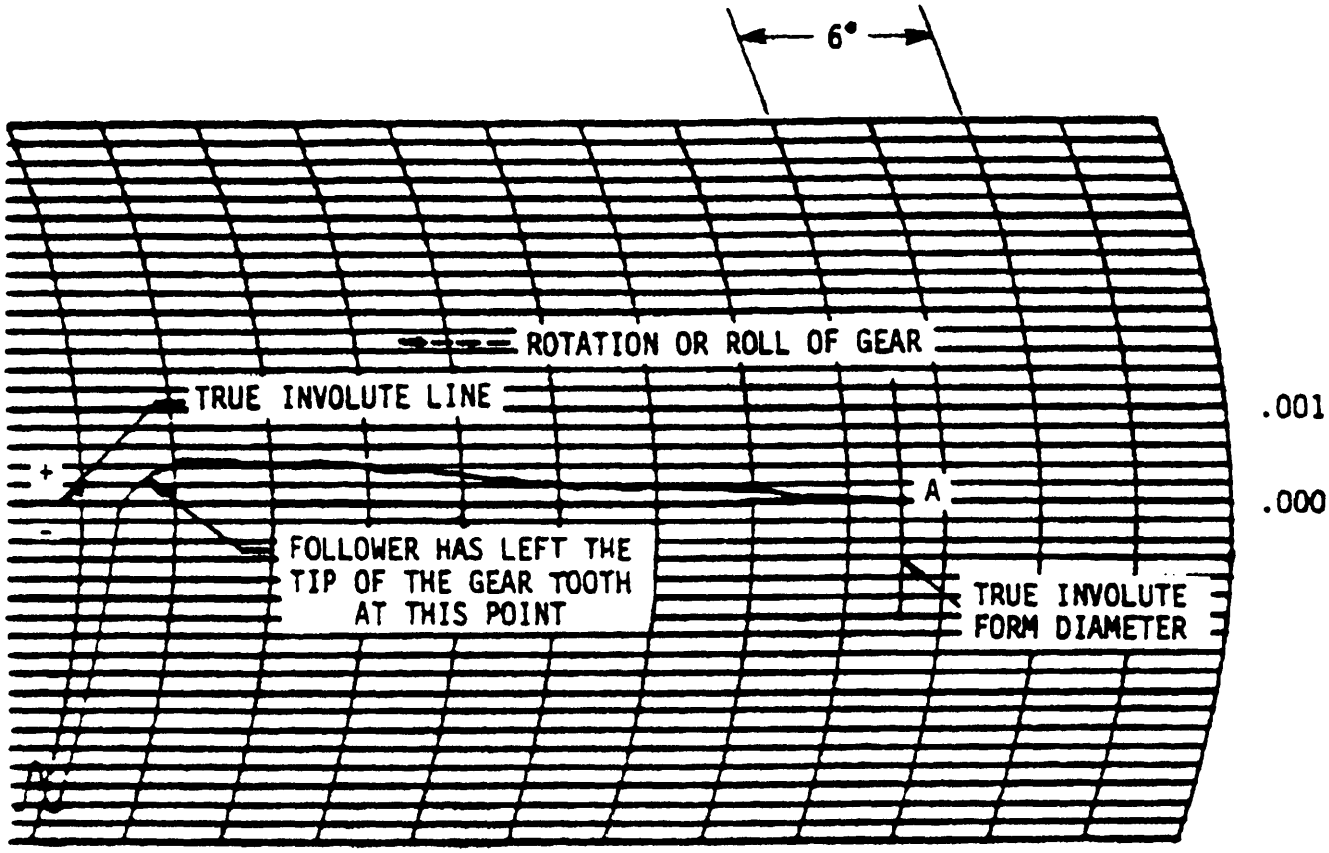


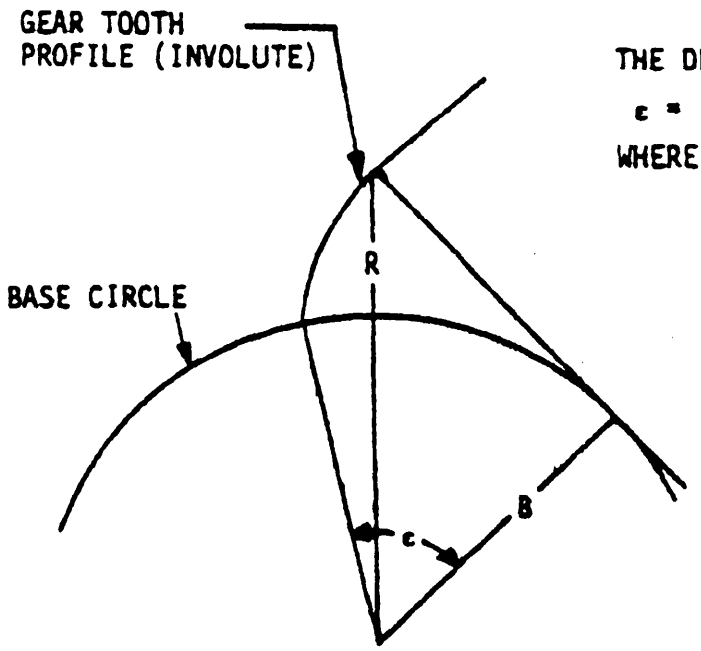
FIGURE 5. Tooth spacing (base pitch) measurement.

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THIS FIGURE REPRESENTS A TRUE RECORDING TAPE AS IT COMES FROM A "FELLOWS" INVOLUTE CHECKER

FIGURE 6. Tooth profile chart.



THE DEGREES OF ROLL

$\varepsilon = 180°/\pi B\sqrt{R^2-B^2}$

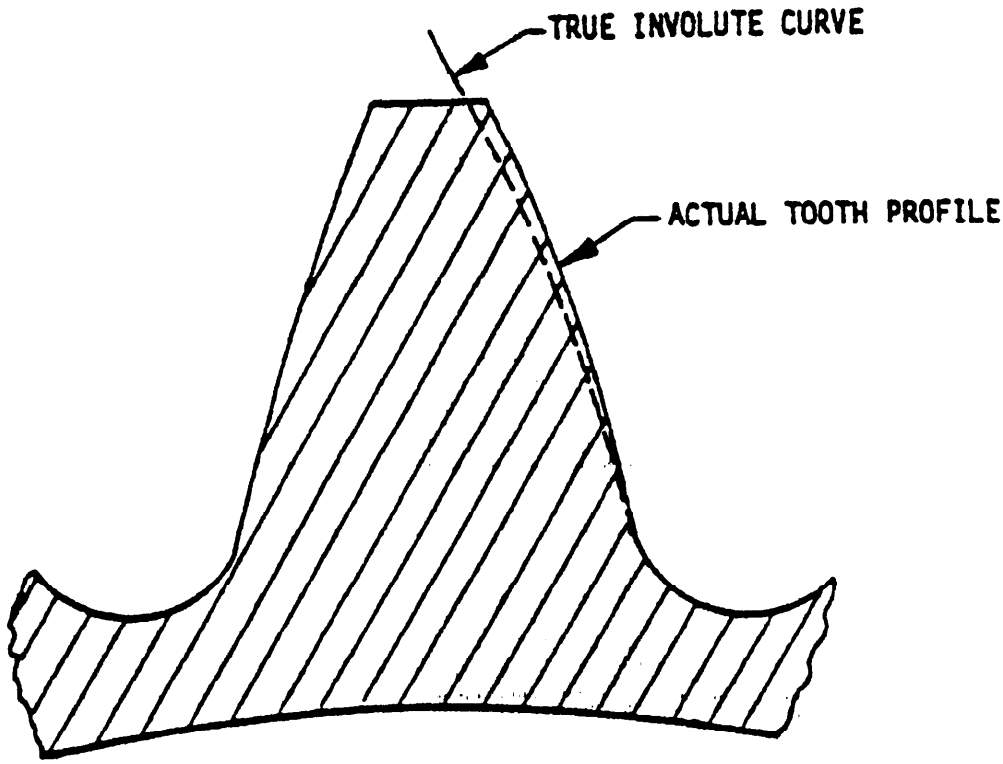
WHERE ε = DEGREES OF ROLL AT A GIVEN RADIUS

B = BASE RADIUS

R = RADIUS AT WHICH THE DEGREES OF ROLL IS TO BE COMPUTED

FIGURE 7. Tooth profile measurement.

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GEAR TOOTH WHICH WAS CHARTED IN FIGURE 6

FIGURE 8. Involute tooth chart.

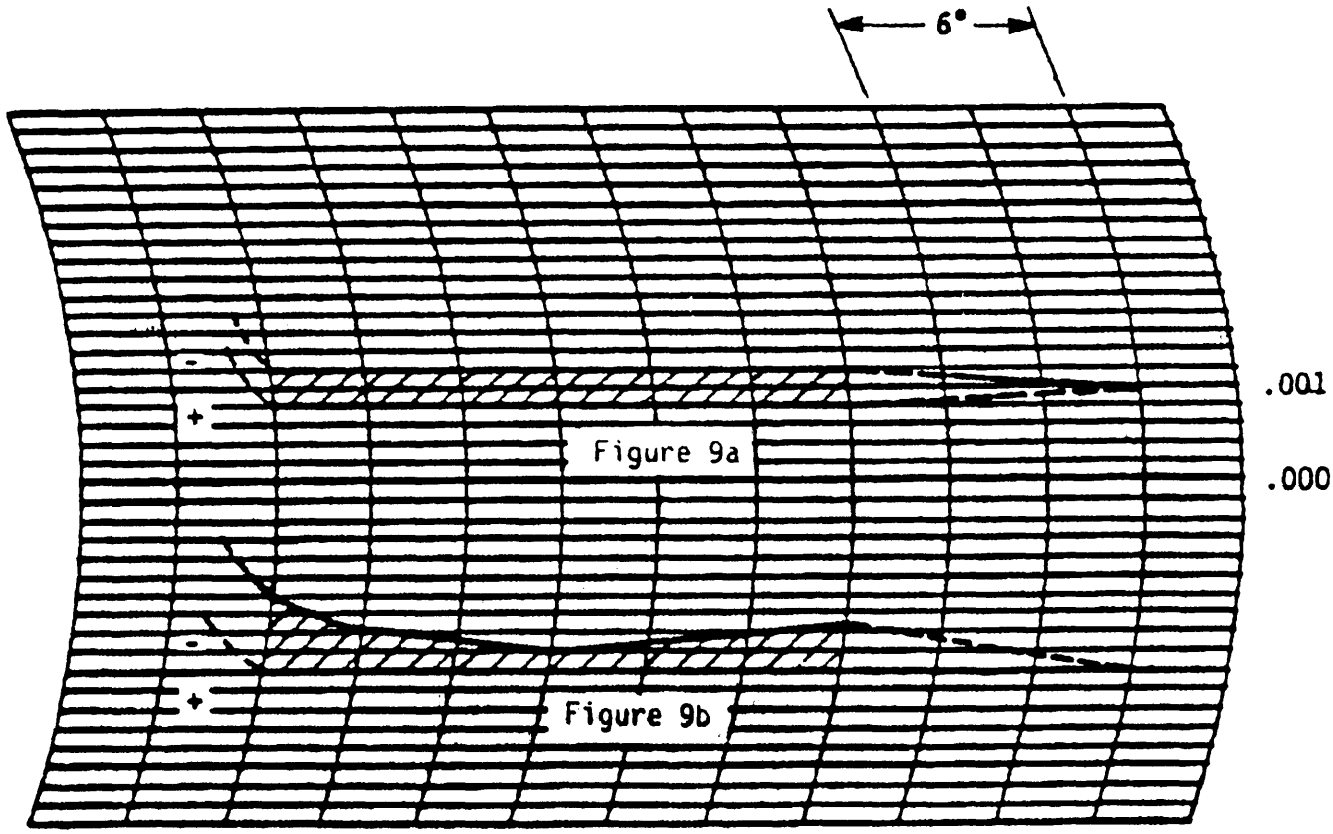


FIGURE 9a. Tooth profile parallel band.

FIGURE 9b. Tooth profile modified band.

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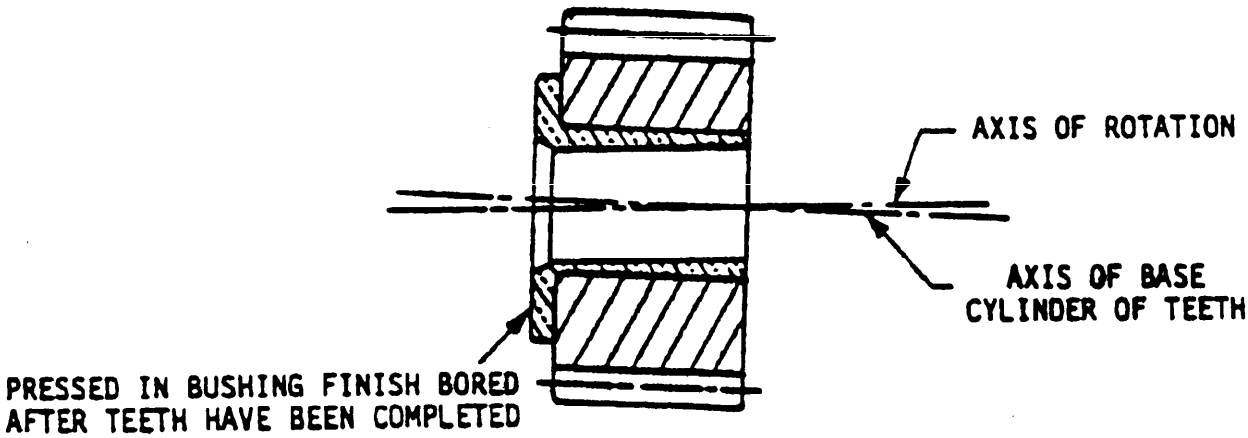


FIGURE 10. Possible deviation between axis rotation, and axis of base cylinder of teeth.

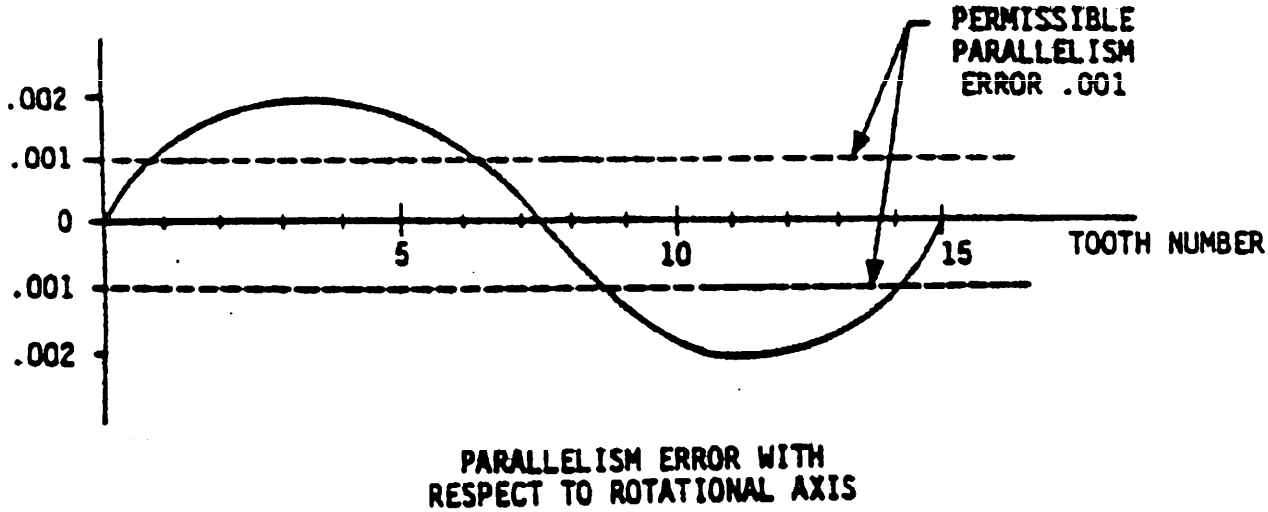


FIGURE 11. Tooth parallelism sine wave.

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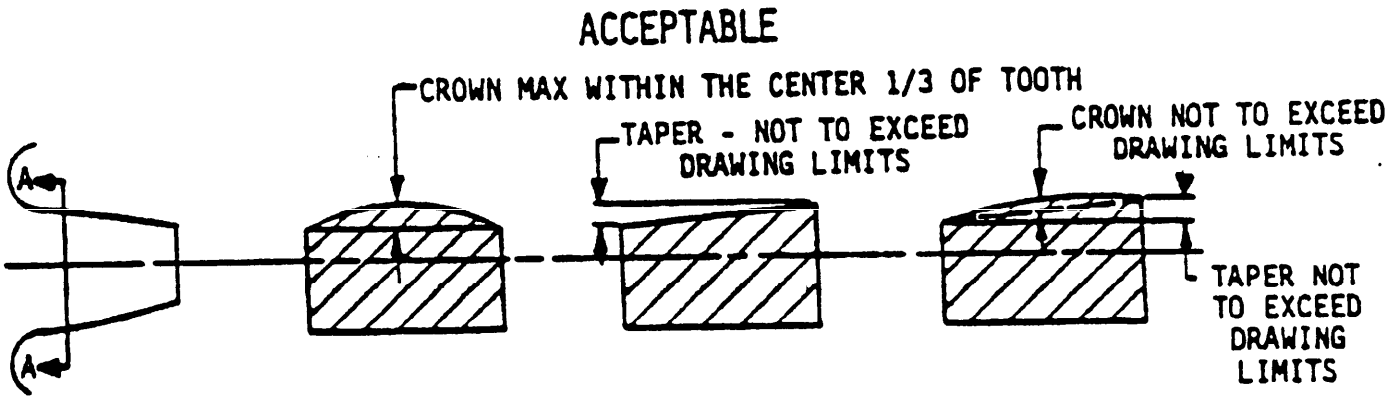


FIGURE 12a. Acceptable tooth surface criteria.

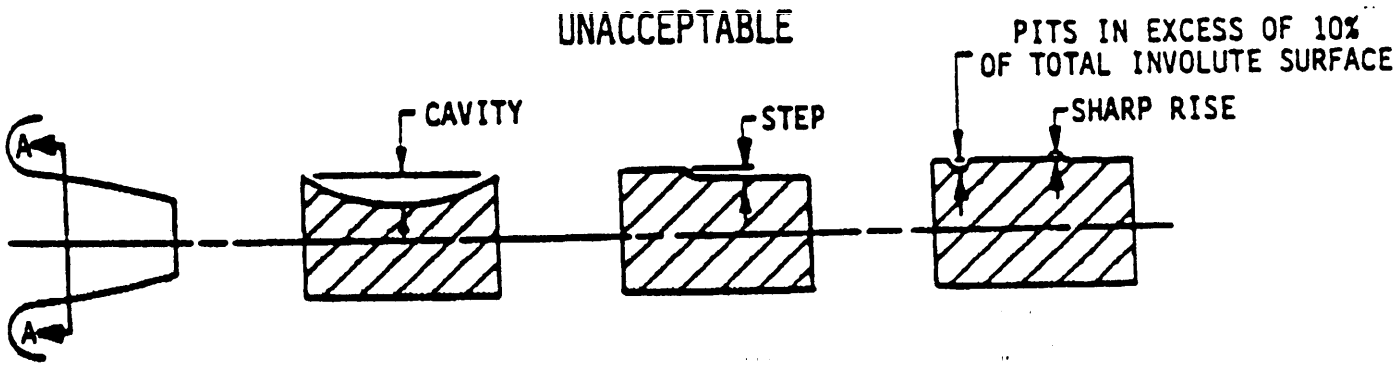


FIGURE 12b. Unacceptable tooth surface criteria.

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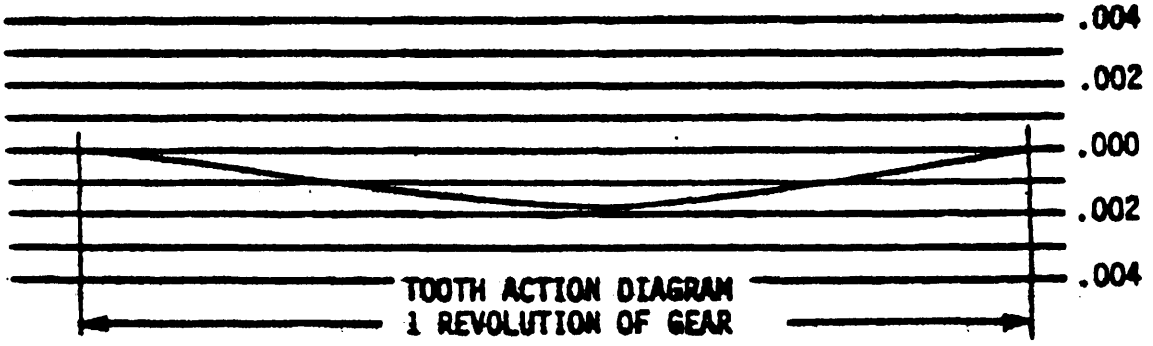


FIGURE 13a. Runout and tooth action error registering eccentricity.

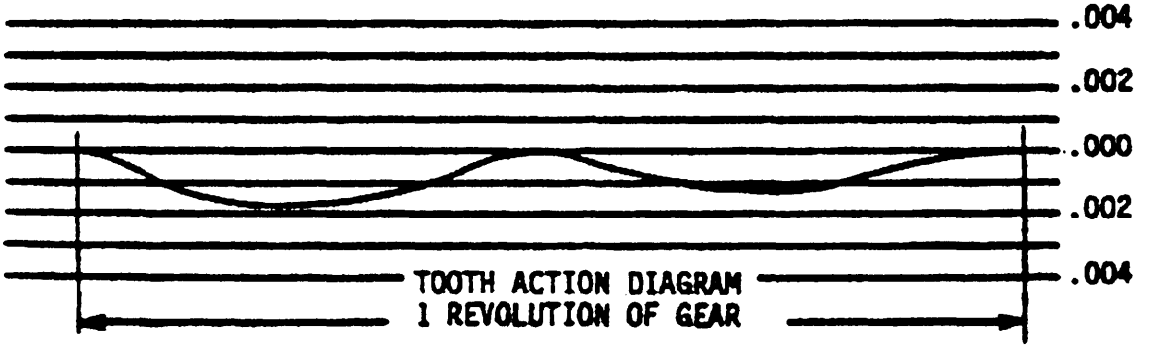


FIGURE 13b. Runout and tooth action error registering out of roundness.

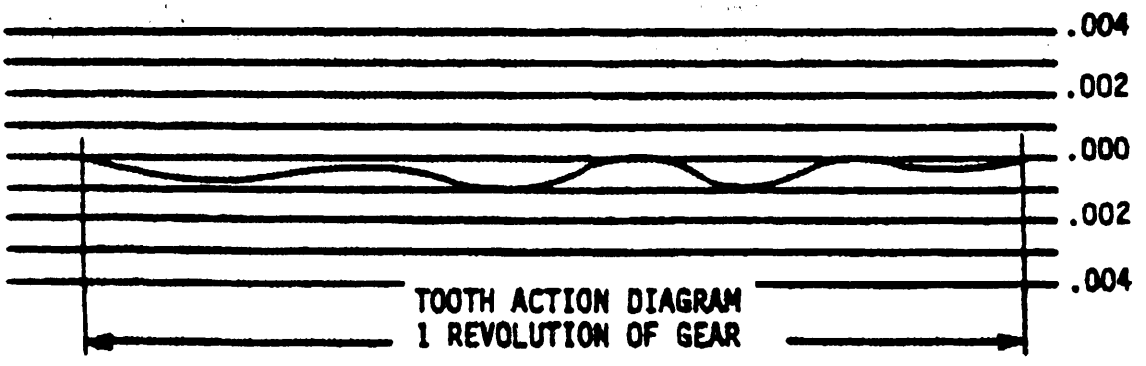


FIGURE 13c. Runout error indicating scallops.

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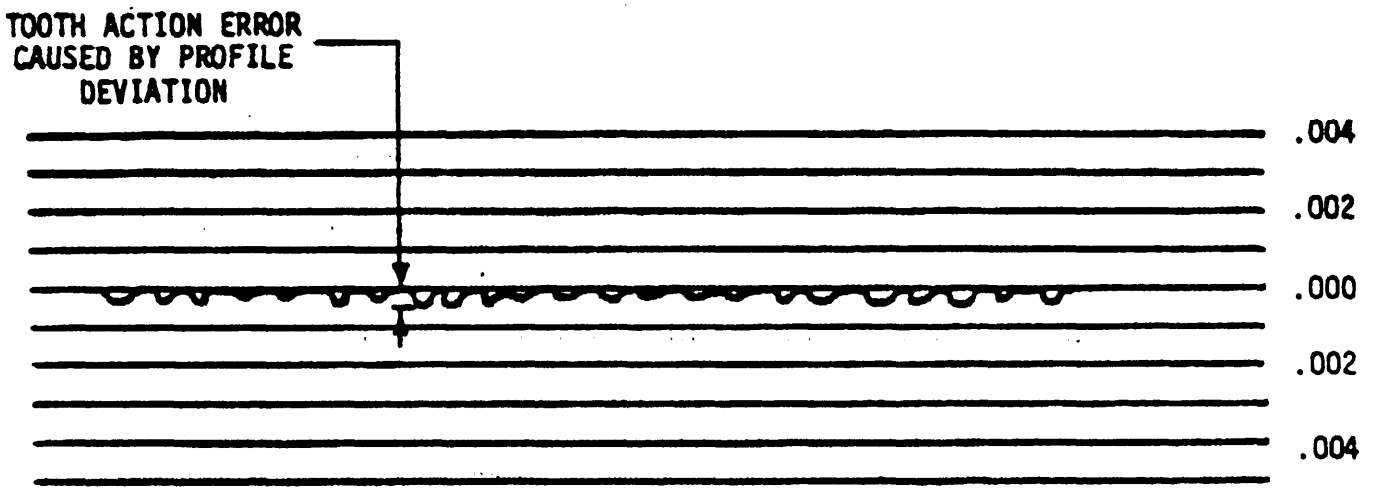


FIGURE 14a. Tooth action error registering profile deviations.

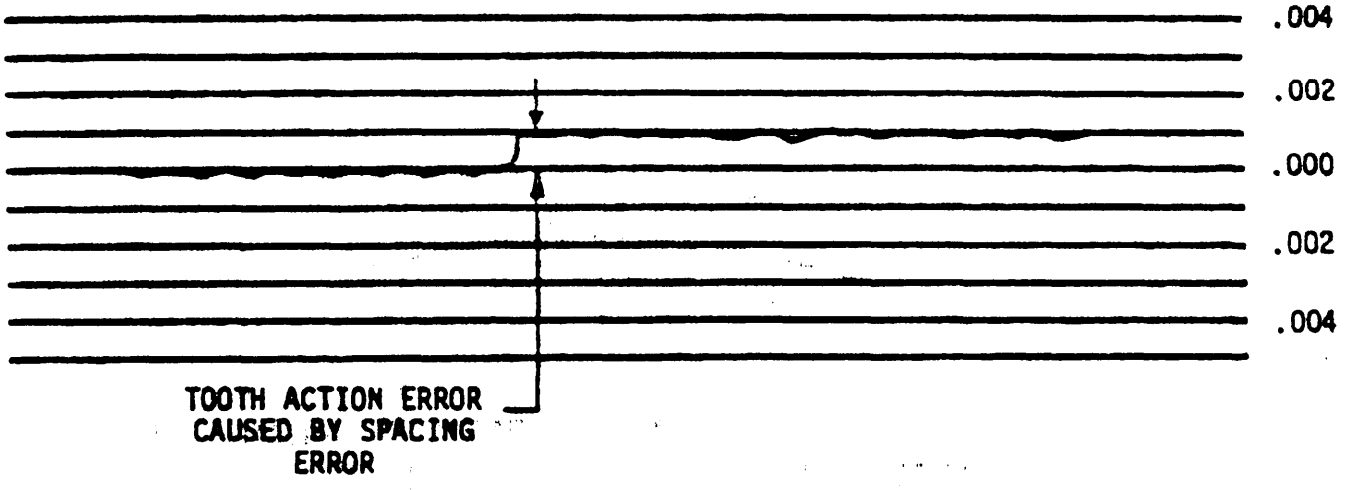


FIGURE 14b. Tooth action error registering spacing deficiencies.

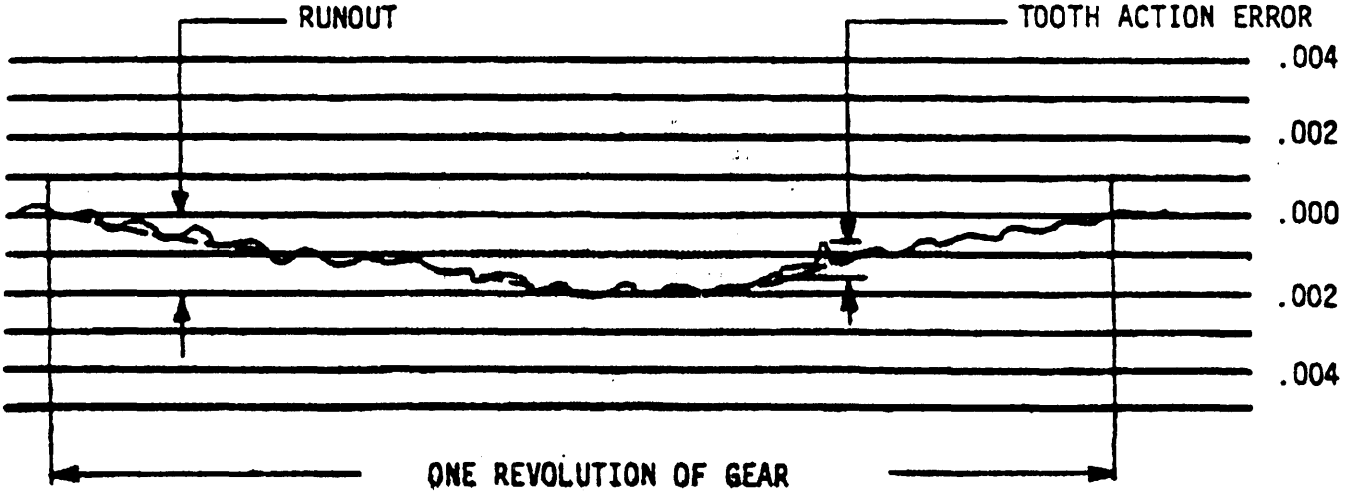


FIGURE 15. Tooth error registering combined runout and tooth action error.

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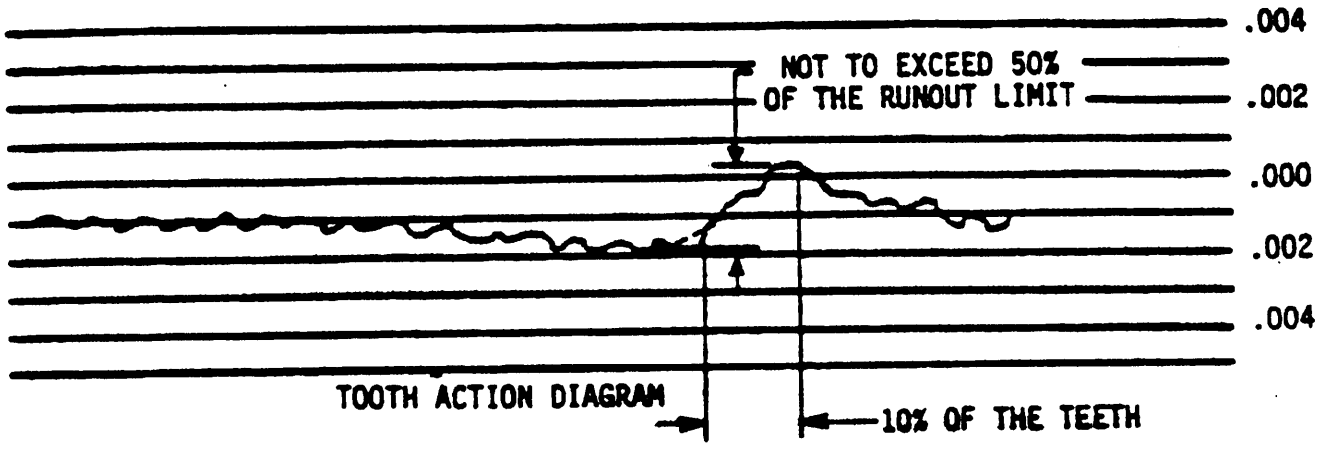


FIGURE 16. Tooth action tolerances.

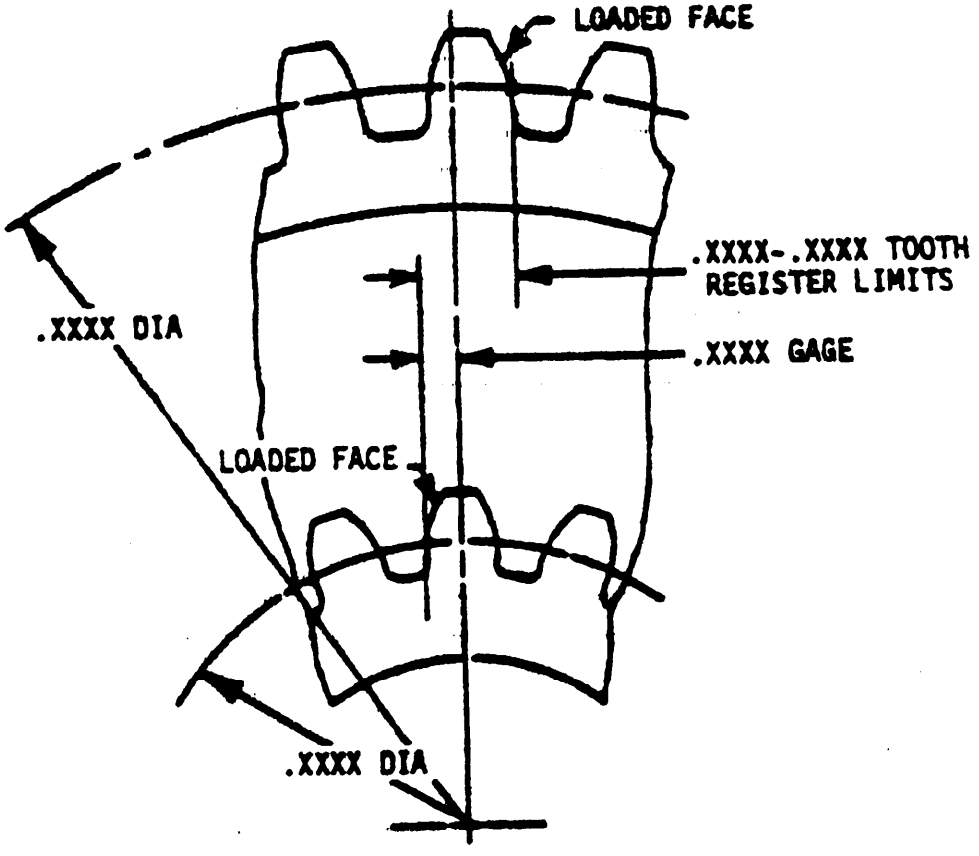


FIGURE 17. Tooth register measurement.

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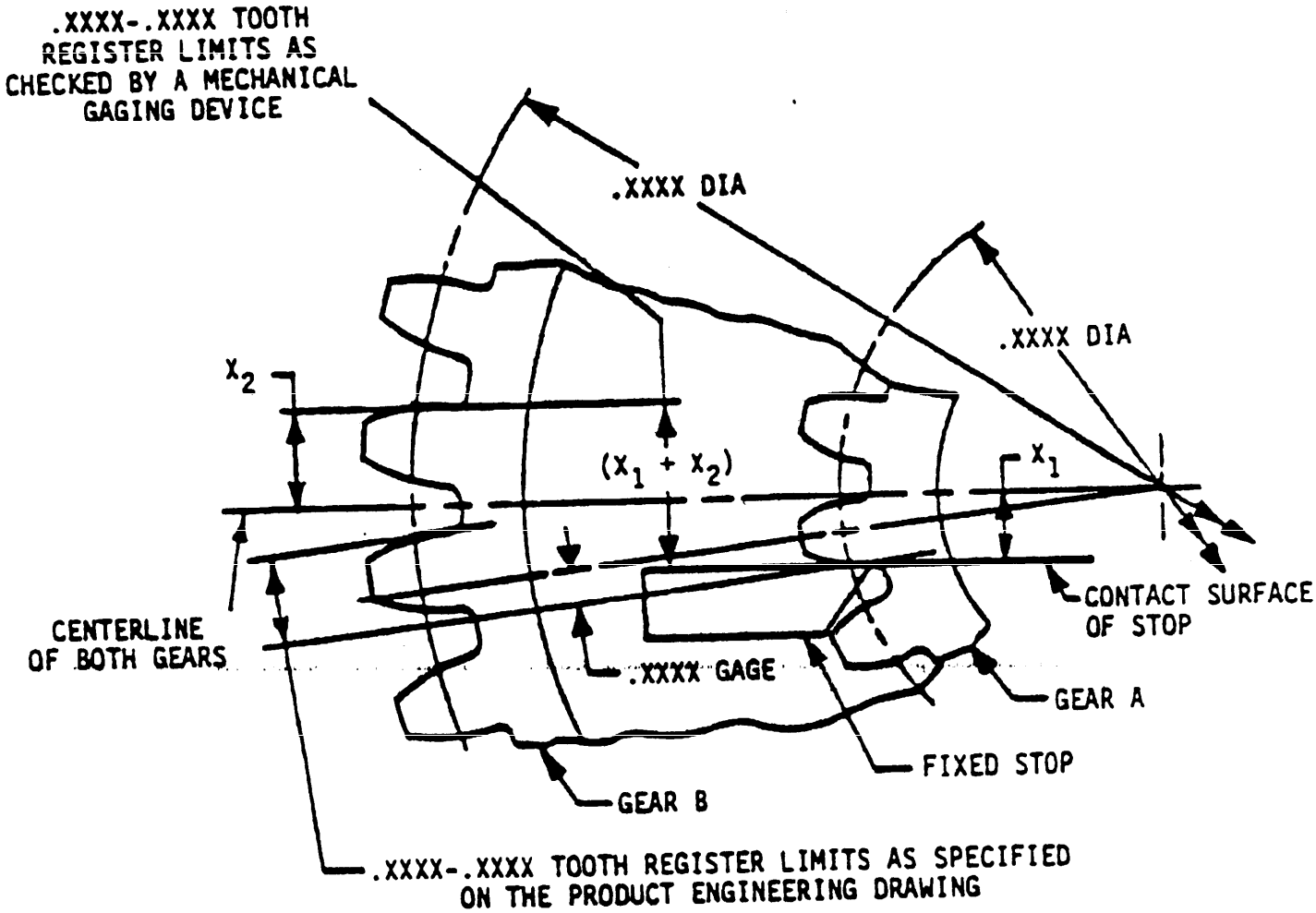


FIGURE 18. Tooth register limit as checked by a mechanical gaging device.

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5.3 Bevel gears.

5.3.1 Applicability. This section provides the methods of controlling and measuring the requirements for acceptance of straight, zerol, and spiral bevel gears when the control is based upon the use of master gears. The manufacture of bevel gears, particularly precision and those requiring utmost quietness in operation, depends vitally on a careful inspection procedure. Because of the complex configurations in bevel gears and the need for conjugate tooth action, it is necessary that diligent inspection be carried out through the whole manufacturing process. In addition, this document covers that portion of the developmental operation which directly affects assembly and inspection procedures.

5.3.2 Classification. Bevel gears shall be one of the types shown in table V.

TABLE V. Bevel gear types.

Item	Bevel		Hypoid	
Gear type:	Straight	Coniflex	Spiral	Zerol
Tooth form:	Octoid Formate	Spherical Helixform	Revacycle Duplex helical	Involute

5.3.3 Development gears. The first step in the manufacture of bevel gears is to produce a trial or development pair of gears which pass inspection and are approved for operation in the final mounting. Following approval several sets of master control gears are produced which duplicate the "developed pair". These hardened control gears are used in the testing machines to control quality at all stages of manufacture. Figure 19 illustrates the dimensional nomenclature pertinent to bevel gear characteristics.

5.3.3.1 Straight (coniflex type), zerol and spiral bevel gears. See figure 20 for bevel gear tooth forms.

5.3.3.1.1 Determining optimum tooth bearing. The development procedure for these gears shall include determination of the optimum tooth bearing with both members of a pair positioned at their normal operating distances and zero offset and run together under a light brake load. Values for normal operating distances are given in the bevel gear data of each bevel gear drawing. The optimum tooth bearing for the above types of gears is one meeting the following requirements.

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5.3.3.1.1.1 Bearing area location. The bearing area shall be slightly nearer to the toe than to the heel and shall be centrally located between the top land and start of fillet. Any deviation from this central location shall be slightly high on the pinion and slightly low on the gear. Figure 21 shows the type of bearing desired.

5.3.3.1.1.2 Bearing area size. The size of the bearing area shall be large as possible without causing it to extend to any of the tooth edges for any extreme combinations of operating conditions. These combinations shall be taken in such a manner as to include vertical offset variations and the operating distance variations for both members.

5.3.3.1.1.3 Bearing area shape. The bearing area shall be essentially of the shape shown in figure 21. It shall have a smooth outline with no abrupt changes and shall be uniform from tooth to tooth. In certain specific cases, it will be necessary to define the optimum pattern more thoroughly.

5.3.3.1.1.4 Extent of bearing area. The tooth bearing area shall not extend to any of the tooth edges at any of the extreme combinations of the following variables taken together:

a. Operating distance of master gear.
b. Operating distance of mating master gear.
c. Vertical offset.

Vertical offset does not refer to vertical with respect to the orientation of the gears in their final mountings. It refers only to the sense of movement in the testing machine. In addition, the value given on each bevel gear drawing is the total value for the set rather than an individual figure for one gear.

5.3.3.1.1.5 Establishing and approving tooth bearing patterns. The preceding requirements represent the optimum tooth bearing patterns developed under ideal conditions. Various factors such as gear tooth geometry, operating conditions and methods of manufacture may make it necessary to depart from these requirements. The cognizant engineering activity shall be responsible for establishing approval of these tooth bearing patterns.

5.3.3.1.2 Developed gear set noise. The developed gear set shall run together smoothly without unusual or excessive noise when checked in the bevel gear testing machine.

5.3.3.1.3 Drawing limits. The developed gear set shall conform to drawing limits for backlash, runout, tooth spacing and location of top land and back face.

5.3.3.1.4 Tooth bearing diagrams. Once having established the optimum tooth bearing, it is necessary to run the set at their respective reference mounting distances and to observe the resulting tooth bearing. A diagram shall be made of both patterns, namely:

a. With both members at their respective normal operating distances and zero offset.

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b. With both members at their respective reference mounting distances and zero offset.

These diagrams shall be approved by the cognizant engineering activity. One set of diagrams shall be included in the process sheets for each gear. In addition, inspection and assembly shall each be furnished duplicate copies of the diagrams.

5.3.3.2 Straight bevel gears not of the coniflex type.

5.3.3.2.1 Development procedure. The development procedure for these gears shall include determination of the optimum tooth bearing with both members at their respective reference mounting distances and zero offset. Values for reference mounting distances are given on every bevel gear drawing. The optimum distances are given on every bevel gear drawing. The optimum tooth bearing for the above gear type is defined to be one meeting the following requirements.

5.3.3.2.1.1 Bearing area. The bearing area shall be substantially full length. It should begin at the small end of the tooth and extend about three-quarters length of the tooth. Figure 22 gives the desired bearing size and location. The bearing is higher on the pinion than on the gear tooth. The development of these gears is to the fixed reference mounting distances and no variations in mounting conditions need be considered.

5.3.3.2.1.2 Shape. The bearing area shall be essentially of the shape shown in figure 22. It shall have a smooth outline with no abrupt changes and shall be uniform from tooth to tooth.

5.3.3.2.1.3 Pattern approval. The optimum pattern established for each set of these gears shall be approved by the cognizant engineering activity.

5.3.3.2.2 Developed gear set. The developed gear set shall run together smoothly and without unusual or excessive noise, when checked in the bevel gear testing machine.

5.3.3.2.3 Check drawing requirements. The developed gear set shall conform to drawing limits for backlash, runout, tooth spacing, and location of top land and back face.

5.3.3.2.4 Prepare diagram. Once having established the optimum tooth bearing with both members at their respective reference mounting distances, a diagram shall be made of that pattern. One diagram shall be included in the process sheets for each gear. In addition, inspection and assembly shall each be furnished duplicate copies of the diagram.

5.3.4 Production gears.

5.3.4.1 Tooth bearing test for straight (coniflex type), zerol and sprial bevel gears.

5.3.4.1.1 Development procedure. The development procedures establish the desired tooth bearing for the following two mounting conditions.

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a. Both members at normal operating distance and zero offset.
b. Both members at reference mounting distance and zero offset.

5.3.4.1.2 Check tooth bearing patterns. It is required that each gear have its tooth bearing checked under these mounting conditions when meshing with its mating production control gear. The bearing patterns obtained shall closely duplicate the established standards of acceptability for production control gears (see figure 23).

5.3.4.1.3 Bearing pattern uniformity. It is required that the tooth bearing be uniform from tooth to tooth. If there is non-uniformity of such a degree to make acceptance or rejection questionable, it is required that both the runout and the tooth spacing be checked. Should these two values be within drawing limits, the gear may be accepted.

5.3.4.1.4 Running noise. Excessive running noise or rough action during the bearing test shall be considered sufficient cause for rejection of the gear. When the noise or roughness is such that acceptance or rejection would be questionable it is required that both runout and tooth spacing be checked. Should these two values be within drawing limits the gear may be accepted.

5.3.4.2 Tooth bearing test for straight bevel gears not of the coniflex type.

5.3.4.2.1 Development procedure. The development procedure establishes the desired tooth bearing with both members at their respective reference mounting distances and zero offset. It is required that each production gear have its tooth bearing checked under these same mounting conditions when meshing with its mating production control gear. The bearings shall closely duplicate those established in development. It is not required that these gears be checked at any combination of mounting variations other than the one combination specified, namely:

a. Zero veritcal offset.
b. Both production gear and mating production control gear at the respective reference mounting distances for the production set.

5.3.4.2.2 Tooth bearing uniformity. It is required that the tooth bearing be uniform from tooth to tooth. If there is non-uniformity to make acceptance or rejection questionable, it is required that both the runout and tooth spacing be checked. Should these two values be within drawing limits, the gear may be accepted.

5.3.4.2.3 Excessive running noise. Excessive running noise or rough action during the bearing test shall be considered sufficient cause for rejection of the gear. When the noise or roughness is such that the acceptance or rejection would be questionable, it is required that both runout and tooth spacing be checked. Should these two values be within drawing limits the gear may be accepted.

5.3.4.3 Other. The acceptable limits for the following characteristics are given on the drawing for each specific bevel gear:

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a. Location of top land and back face.
b. Backlash.
c. Runout.
d. Tooth spacing.

NOTE: Because of the length of time required to inspect a gear for runout or tooth spacing, they are only required when any of the following is detected in the gear tests:

1. Non-uniform tooth bearing from tooth to tooth.
2. Questionable noise.
3. Questionable degree of roughness in operation.

5.3.4.4 Common gear. When one gear mates with two or more gears at the same time and two or more meshes have a different developed tooth bearing pattern, the common gear shall have the proper developed tooth bearing pattern with all its mating production control gears. The common gear need only be checked with one of its mating production control gears to save time, but shall conform to the above requirement.

5.3.5 Master control gear.

5.3.5.1 Model master control gears. The model master control gears shall conform to the same requirements as the production gear specified in 5.3.4 with the exceptions cited in paragraphs 5.3.5.1.1 through 5.3.5.1.4.

5.3.5.1.1 Tooth bearing test. The model master control gear shall be run with its mating model master control gear in the tooth bearing test.

5.3.5.1.2 Marking master control gears. The tooth thickness of the model master shall be made equal to the theoretical tooth size minus a portion of whatever backlash manufacturing may deem necessary or suitable to them. Usually one-half of the total required backlash (with both model masters at minimum operating distance and zero offset) will be cut into each of the gears. The amount cut into each gear shall be clearly and permanently etched on the model master gear to the nearest ten thousandth of an inch. It is emphasized that this etched value is not the total backlash for the set of model masters, but is an individual figure for the one member. The total backlash for the set (with both model masters at minimum operating distance and zero offset) is therefore equal to the sum of the etched values for the two model masters.

5.3.5.1.3 Runout. The runout of any master control gear shall not exceed 0.0005 inch.

5.3.5.1.4 Use of master control gears. Master control gears are used only to check the inspection control gears.

5.3.5.2 Inspection control gears.

5.3.5.2.1 Inspection control gears. Inspection control gears shall conform to the same requirements as the master control gear specified in 5.3.5.

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5.3.5.2.2 Check for wear. Inspection control gears shall be checked periodically for wear. Should the total backlash for the set (with both members at minimum operating distance and zero offset) exceed the sum of the etched values by 0.0005 inch or more, the etched values shall be corrected accordingly.

5.3.5.3 Production control gears. The production control gears shall conform to the same requirements as the production gear specified in 5.3.4, with the exceptions cited in paragraphs 5.3.5.3.1 through 5.3.5.3.4.

5.3.5.3.1 Tooth bearing test. The basic production control gear shall be run with the mating production control gear in the tooth bearing test.

5.3.5.3.2 Backlash. The tooth thickness of the production control gear shall be controlled by limiting its backlash when in mesh with its mating inspection control gear (with both members at their respective minimum operating distance and zero offset). This backlash shall be B + .001, -.000 where B is that portion of the total backlash between the set of inspection control gears that have been etched on the mating inspection control gear.

5.3.5.3.3 Production control gear. Each production control gear shall be periodically checked for wear. This shall be done by measuring the backlash when in mesh with its mating inspection control gear (both members at their respective minimum operating distances and zero offset). Should the measured backlash ever exceed the value B +0.001, where B is the value etched on the inspection control gear, the inspection control gear shall be inspected to determine whether the excess backlash is due to any permissible wear of the inspection control gear. If the value determined by the inspection of the inspection control gear coincides with the etched value, the production control gear is no longer acceptable for use. Should the inspection control gear show any amount of permissible wear, both the upper and lower limits of the backlash between the production and the inspection control gear (with both members at their respective minimum operating distances and zero offset) shall be increased by an amount equal to the amount of wear of the inspection control gear. If the backlash between the production control gear and mating inspection control gear falls within this new range of backlash, then the production control gear is acceptable for use. If the backlash between the production control and inspection control gear does not fall within this corrected range, the production control gear is not acceptable for use.

5.3.5.3.4 Runout. The runout of any basic master gear shall not exceed 0.0005 inch.

5.3.5.4 Mandatory. When one gear mates with two or more gears at the same time and two or more meshes have a different developed tooth bearing pattern, control gears shall be established for all gears.

5.3.5.5 Hardening. All gears that have been accepted as meeting the requirements of this standard and have been run in with the green production control test shall be surface hardened in accordance with the bevel gear drawing requirements.

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5.3.5.6 Final test. A final test is usually made on each pair after hardening, or in the case of curved-tooth bevel gears, after subsequent lapping or grinding. Where adjustability is provided in the final mounting, the testing machine provides a rapid and easy means of determining, for each pair of gears, the mounting distances at which they operate best. In this case, the gears are matched in sets and usually wired together. It is common to mark the measured "mounting distance" and "backlash" values on the outer end of the teeth.

5.3.5.7 Finish. Surface finish of the gears shall be as specified on applicable part drawings. Gears shall be deburred and sharp edges removed.

5.3.5.8 Marking. Each gear shall have identification markings etched or stamped as specified on applicable drawings, or in accordance with contract requirements. Each bevel gear set shall be serialized and permanently marked as matched sets, and shall be stocked and issued as matched sets.

5.3.6 Test methods.

5.3.6.1 100% inspection. Each bevel gear manufactured shall be matched with the mating gear or pinion and mounted on a bevel gear testing machine provided with axial adjustments for the gear and pinion, and with a vertical offset adjustment for raising or lowering the axis of one member vertically with respect to the other member for the testing of hypoid gears. The following requirements shall be met:

a. Tooth bearing pattern.
b. Noise and smoothness of operation.
c. Location of top land and back face.
d. Backlash.
e. Runout.
f. Tooth spacing.

Gear sets meeting the above requirements shall be serialized and retained as matched sets.

5.3.6.2 Failure. Gear sets not meeting the test requirements in 5.3.6 shall be rejected.

5.3.6.3 Detailed methods.

5.3.6.3.1 Tooth bearings. The test for bevel gears is a check of their operation under normal running conditions in their final mountings. Bevel gear testing machines are designed to provide a running test of the gear under simulated service conditions of running. Controlled axial and vertical movements are provided in the machine in order to simulate service deflections as well as to allow for the mis-positioning of the gears due to housing tolerances, bearing play, etc. In making this test, the production gear and its mating gear are mounted at their respective selected operating distances and run together under a light brake load. Prior to running, the gears are painted with a non-abrasive marking compound. A run of a few seconds is sufficient to rub off the compound in the area of tooth contact. The tooth contact pattern shall meet the standards shown in figures 21, 22 or 23 as applicable.

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the extreme combinations of the ... extend to any of the tooth edges or any o...

5.3.6.3.2 Noise and smoothness of operation. The test for noise and smoothness is a check of gear operation under normal running conditions in the final mountings. This test should be combined with the tooth bearing check described above. While the gears are running in mesh for the tooth bearing test, the operator should listen for any unusual noise types and levels. Any other indications of rough action should also be recorded.

5.3.6.3.3 Location of top land and back face. These surfaces may be checked as shown in either figure 24 (part A or B). However, inspection may develop other methods (for example, comparator methods) so long as they are adequate to insure conformance of the gear to drawing requirements. The drawing for each particular gear furnishes sufficient basic dimensions to properly position and zero the dial indicators (see figure 24, part A) or to properly position the contact points (see figure 24, part B). Figure 25 shows the specific basic dimensions given on the drawing. It should be noted that the basic diameter given is a dimension which is concentric with the X diameter. It is important that the set-up of the checking device be made to satisfy this requirement. Once having accomplished this set up, the method of figure 24, part A, checks the surface locations by direct reading of the dial indicators, while the method of figure 24, part B, makes use of a feeler gage to be inserted between the contact points and the production gear. For either method it is not necessary that surfaces be checked along their entire length. Two points reasonably close to either end (lengthwise) of the particular surface will be sufficient.

NOTE: Diameter X will be specified on the drawing. It will be selected as that diameter upon which the gear is mounted in final assembly.

5.3.6.3.4 Backlash. The measurement of backlash is made in the gear testing machine. The gear being inspected is assembled in mesh with its mating gear. With the gear fixed, the rotation of the production gear is measured using a dial indicator as shown in figure 26. This measurement is taken normal to the active tooth surface at the extreme heel of the gear. This is a suggested method, other methods may be employed so long as they are equivalent to this. Care should be taken to convert all readings to normal backlash should the method employed not indicate this value.

5.3.6.3.5 Runout. On bevel gears the runout tolerance includes both axial and radial runouts (see figure 29). Measuring methods are defined in the following four (4) paragraphs.

5.3.6.3.5.1 Rolling check. Axial runout is measured in a bevel gear testing machine. The gear being inspected is rolled in tight mesh with a master control gear having a maximum runout of 0.0005 inches. The variation in the mounting distance depicted in figure 27 is the axial runout.

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be approved by the cognizant engineering activity. set of these gears shall

5.3.6.3.5.2 Runout bearing variation. Variations in runout may be observed by running the gears in a suitable test machine. The runout is characterized by periodic variation in sound during each revolution, and by the tooth bearing shifting progressively around the gear from heel to toe, and toe to heel. Figure 28 depicts these variations. This test also includes the effects of tooth element variations. Runout is observed by visually checking variation in contact pattern. The actual amount should be determined by either the single or double probe method defined in the following paragraphs.

5.3.6.3.5.2.1 Single probe runout check. Measurements made with a single probe as depicted in figure 29 are used to determine true runout variations when measured perpendicular to the surface of revolution. The surface of revolution of a bevel gear is the pitch cone. A single probe can also be used to isolate axial or radial effects on total runout tolerance.

5.3.6.3.5.2.2 Double probe runout check. As shown in figure 30, this device has two contact points 180 degrees apart, one a fixed stop, the other actuating a dial indicator reading to ten thousandths of an inch. The gear to be checked is mounted on a dead center, held to the vise, and indicator readings are taken on diametrically opposite teeth, with the contacts acting on the corresponding sides of these teeth. The contact points are mounted on a swinging bracket, which can be lifted while the gear is turned from tooth-to-tooth. The change between the largest and smallest readings obtained in one revolution of the gear is a value which is four times the eccentricity (twice the runout) of the pitch circle defined by the sides of the teeth. Thus, this test is a very sensitive test of the eccentricity of a gear. It measures eccentricity by a factor of 4, and tooth spacing errors by a factor of 1, thus the test has only minor uncertainty due to side effects.

5.3.6.3.5.3 Tooth spacing. The spacing of individual teeth of a bevel gear is measured in a Gleason tooth spacing tester or equivalent. This device has two contact points which can be adjusted to contact corresponding sides of adjacent teeth. One contact point is used as a fixed stop, and the other actuates a dial indicator reading to ten-thousandths inch. The two contact points are carried on a swinging bracket which can be lifted out of position and then returned to permit the checking of successive teeth in the gear. The indicator is set to zero for the first pair of teeth. All subsequent readings are therefore relative to the first pair of teeth selected. In interpreting the readings, it is important to note that the readings are not the tooth-to-tooth spacing errors. These errors are the differences between successive readings. The reliability of this check is limited by the sensitivity of the equipment and the skill of the operator. However, experience shows that the slight misinterpretations of the dial indicator readings are as often in the minus direction as they are in the plus direction, thereby cancelling each other when a reasonable number of teeth are checked. To illustrate the method, the readings for a ten tooth gear are given in table VI.

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5.3.6.3.5.4 Hardness check. Hardness checks shall be made to determine conformance to drawing requirements. A minimum of three (3) samples shall be selected from each heat treat lot for determination of effective case depth, core hardness and surface hardness. Samples shall be processed by the same heat treat process procedure used for production parts. Inspection and hardness tests shall be performed in the areas shown in 5.1.

5.3.6.3.5.4.1 Failure. Failure of any one (1) test sample shall be cause for rejection of the entire representative production lot. Items produced subsequent to failure of the test sample shall not be submitted for acceptance until corrective action has eliminated the cause of the failure. Acceptance of heat treat samples subsequent to failure shall not resume until three (3) consecutive samples have successfully met specified test requirements.

5.3.6.3.5.5 Finish. Verify that the surface finishes specified on applicable part drawings conform to specified requirements, and that there are no burrs or sharp edges on any part of the gears.

5.3.6.3.5.6 Marking. Verify that the gears are permanently marked as specified on applicable part drawings and that the bevel gear sets are serialized and maintained as matched sets.

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TABLE VI. Readings in ten-thousandths of an inch.

Column 1 Tooth pair	Column 2 Indicator reading	Column 3 Difference between successive readings of column 2
1	0	
		+2
2	+2	
		+1
3	+3	
		+1
4	+4	
		-2
5	+2	
		-2
6	0	
		+1
7	+1	
		-3
8	-2	
		+2
9	0	
		+3
10	+3	
		-3
1	0	
		SUM 0

Since the gear has a full number of teeth, any spacings which are too large must be offset by others which are too small. Therefore, the sum of all the differences (column 3) must be zero. If uncompensated errors are made in reading the indicator, the sum of column 3 will not equal zero. If the sum of column 3 deviates from zero by more than 0.0002 inch, the check must be repeated. The largest number (regardless of sign) in column 3 is the maximum tooth to tooth spacing error.

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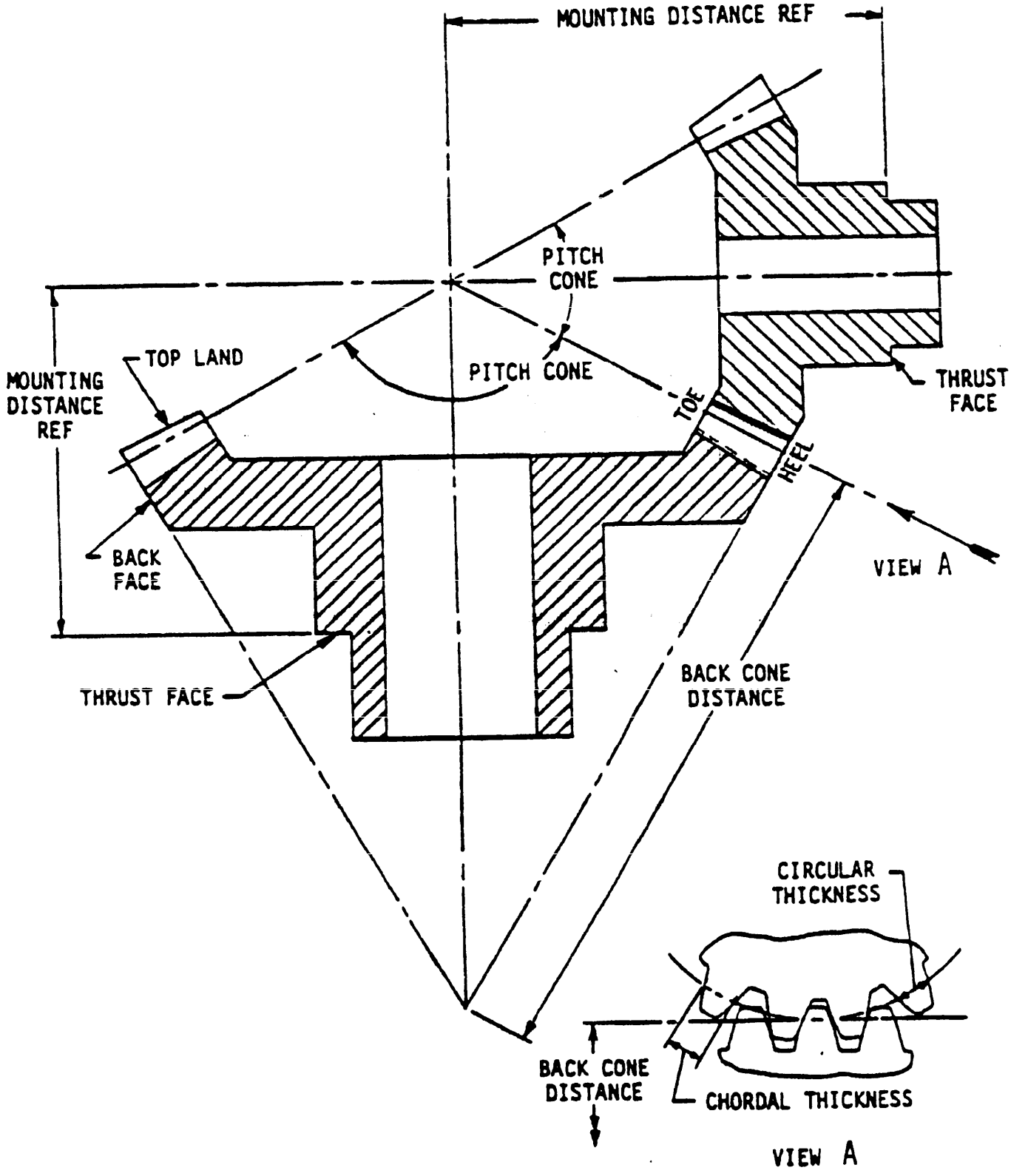
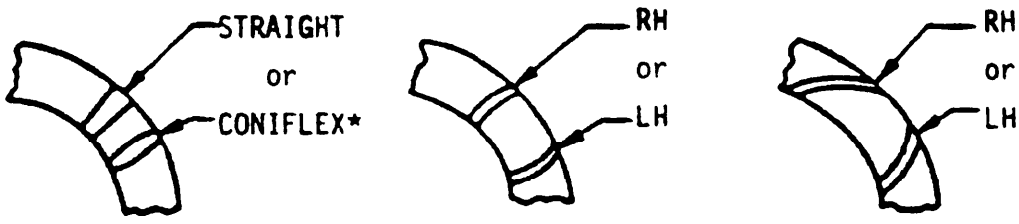


FIGURE 19. Bevel gear nomenclature.

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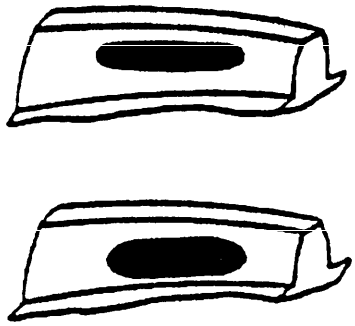


STRAIGHT ZEROL* SPIRAL

*Registered in US Patent Office

FIGURE 20. Bevel gear tooth forms.

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NOTE: Also a typical optimum bearing for coniflex straight, and zerol gear sets.

FIGURE 21. A typical optimum tooth bearing for a spiral bevel gear set.

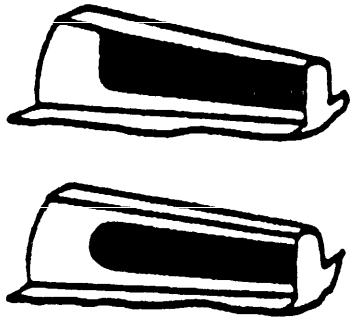
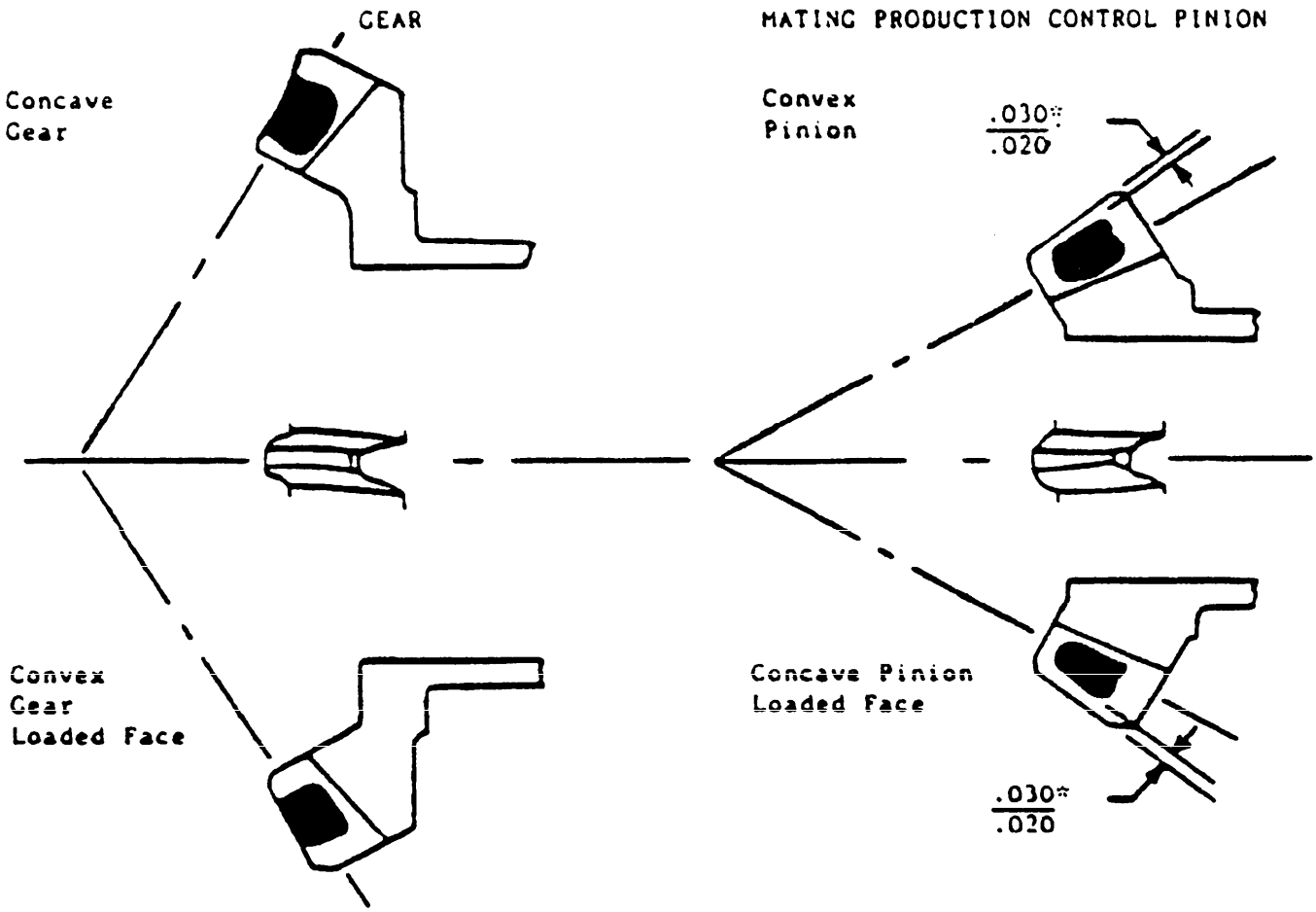


FIGURE 22. Optimum tooth bearing for straight bevel gear not of the Coniflex type.

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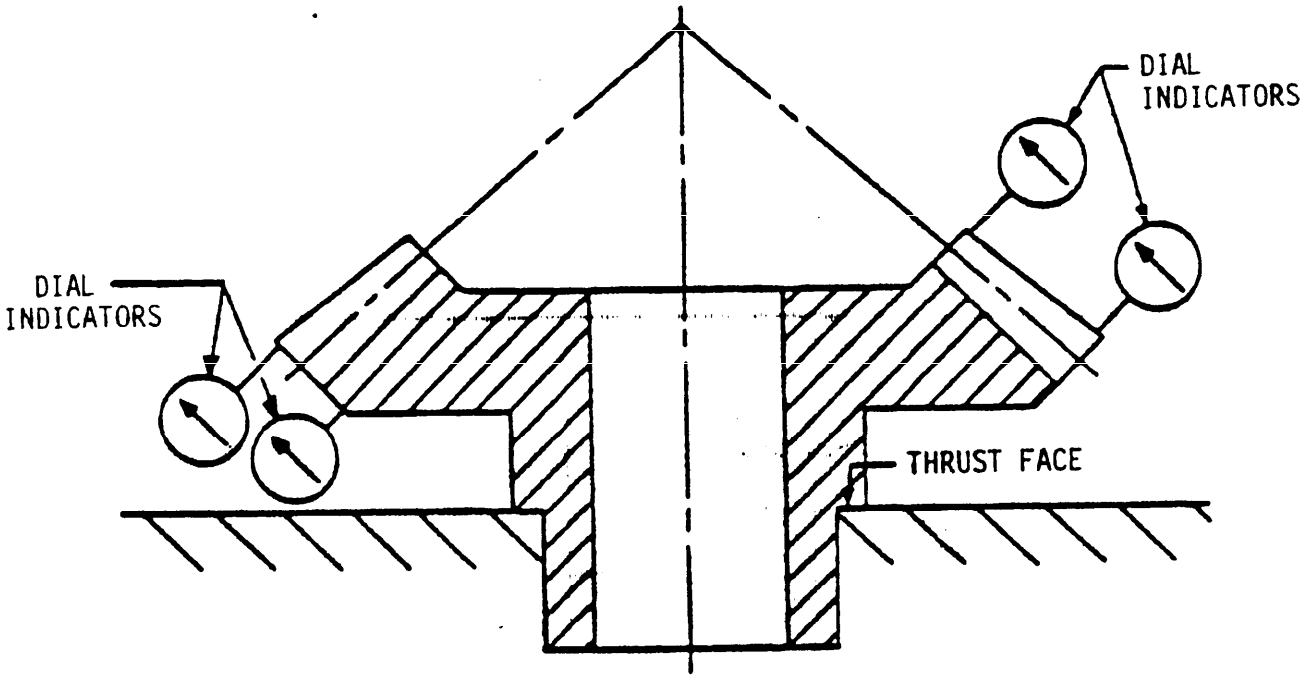


Scale ≈ 2X size
*Full size dimensions

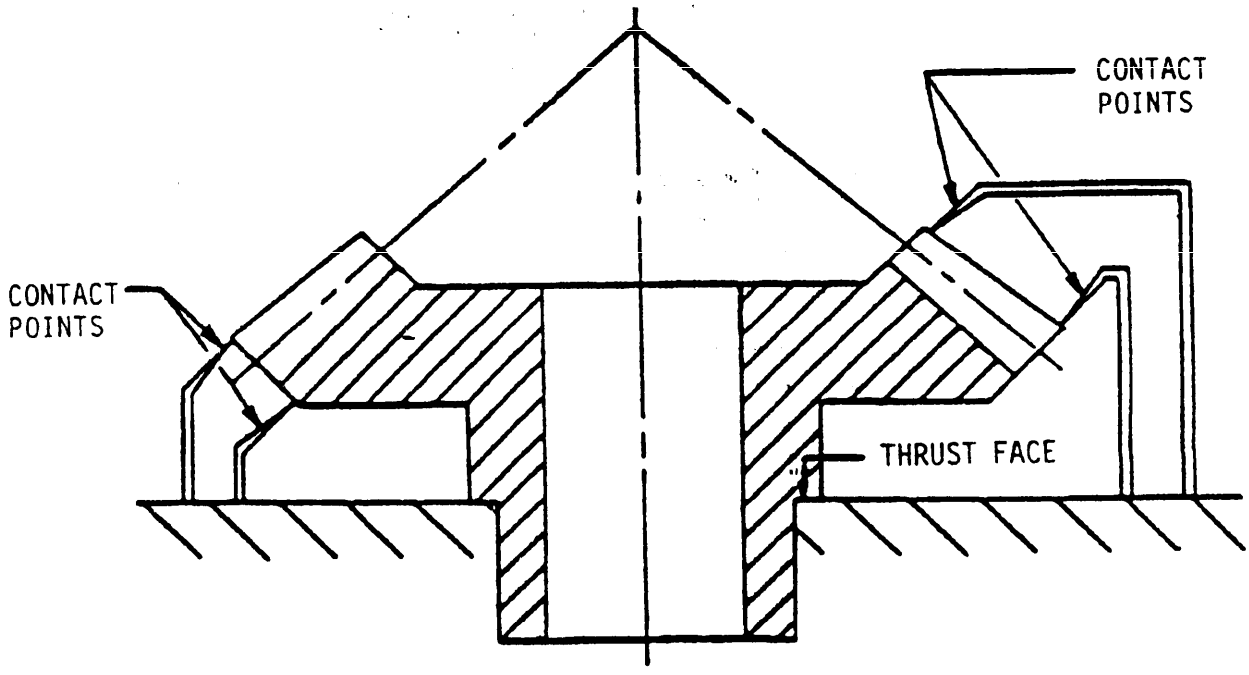
Note: In addition to meeting the pattern diagram, the tooth bearing pattern shall also show the .020 to .030 inch band at the top land edge of the mating basic master pinion at the normal operating distance.

FIGURE 23. Tooth bearing patterns.

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PART A



PART B

FIGURE 24. Two methods for checking the location of top land and back face.

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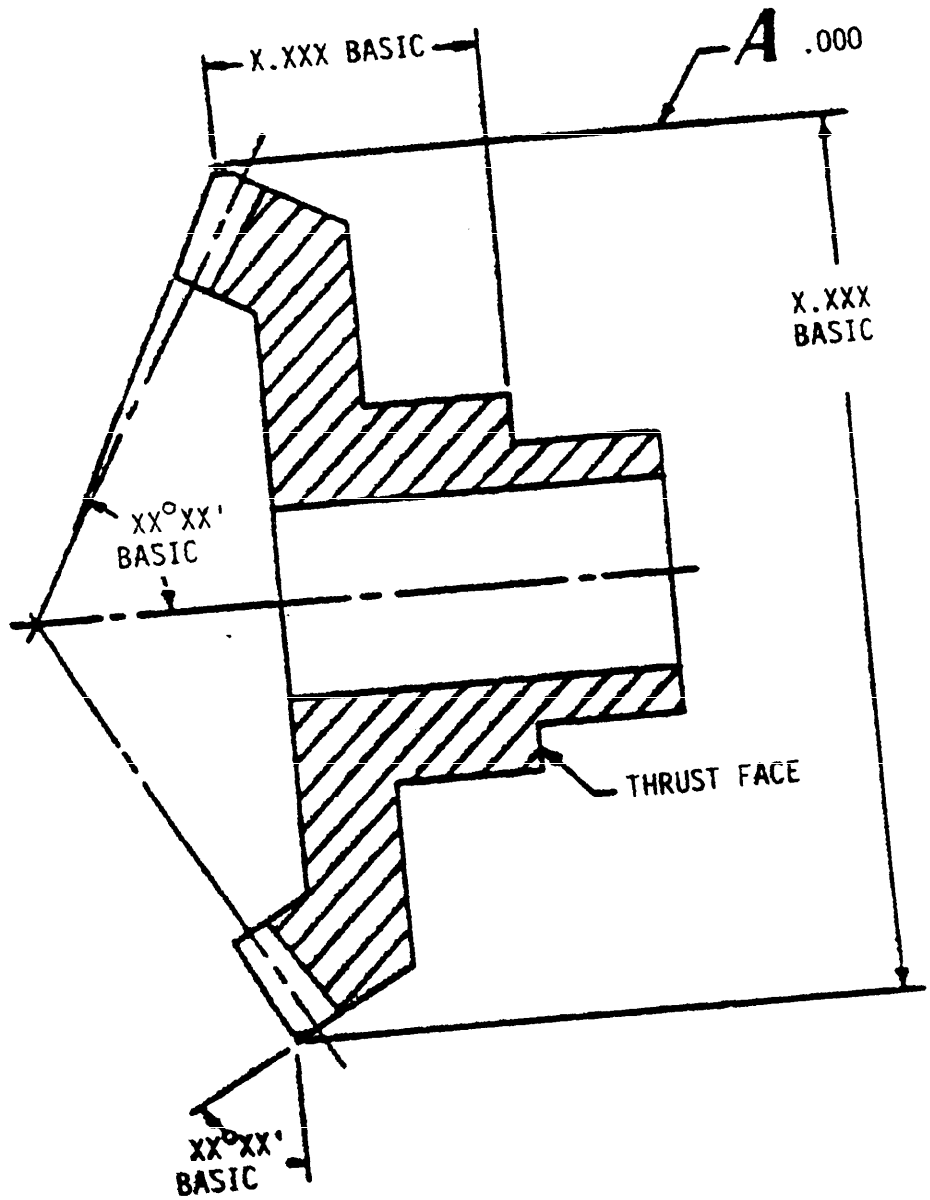


FIGURE 25. Basic dimensions furnished on each bevel gear drawing.

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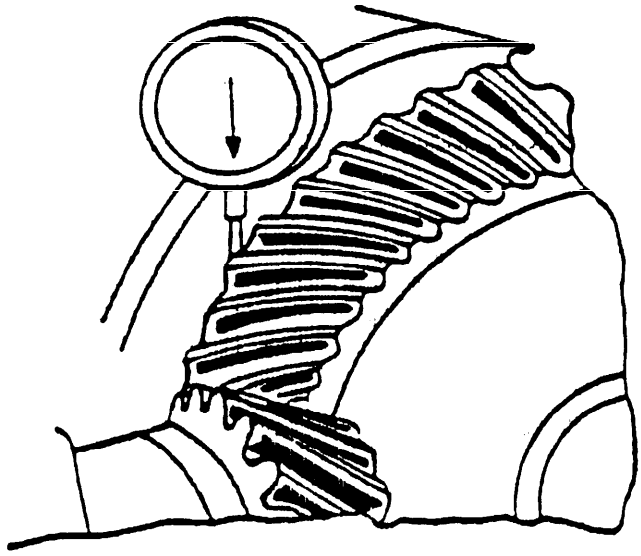


FIGURE 26. Backlash measurement.

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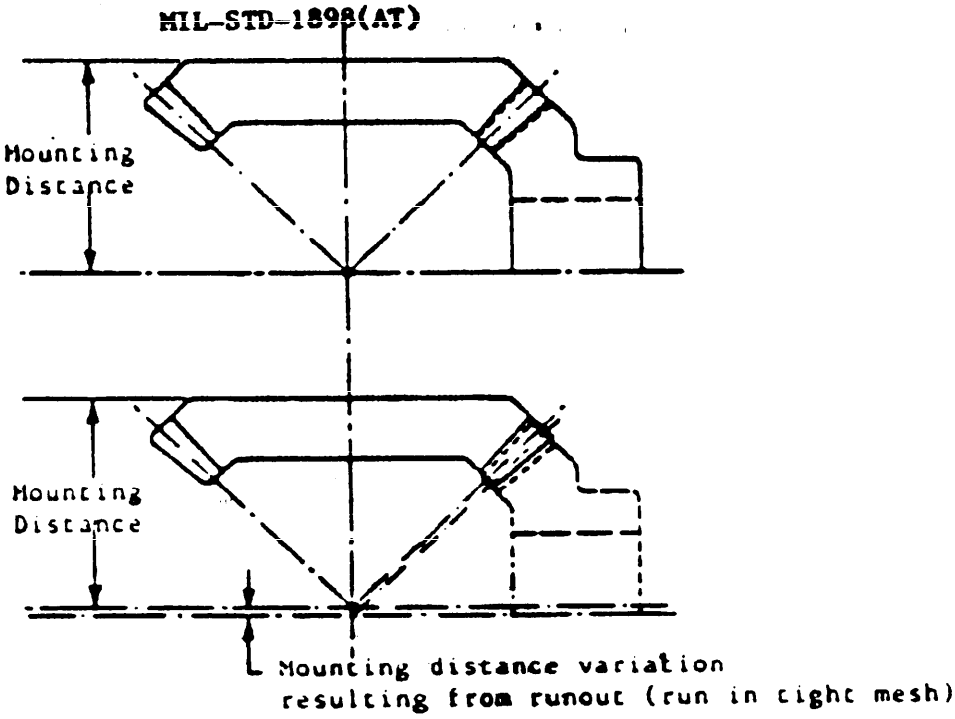
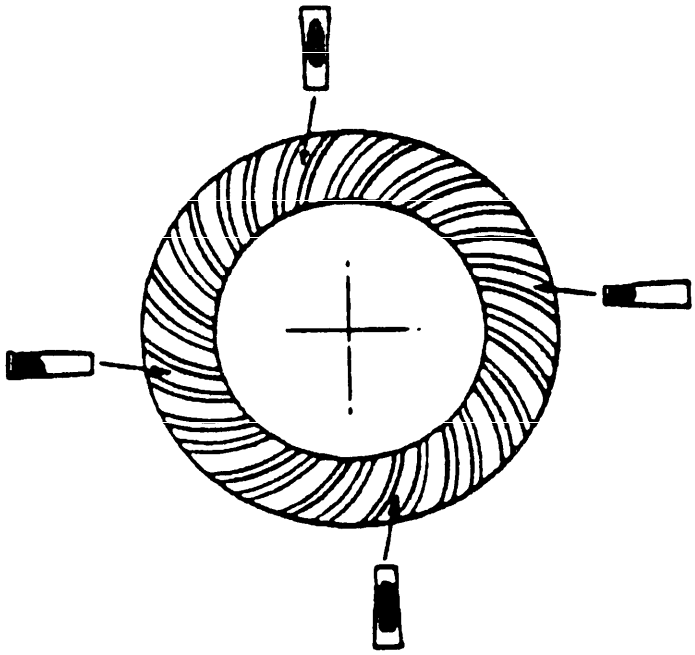


FIGURE 27. Rolling check.



Shifting of tooth contact shows presence of runout. Sound variation also characterizes the existence of runout.

FIGURE 28. Runout bearing variation.

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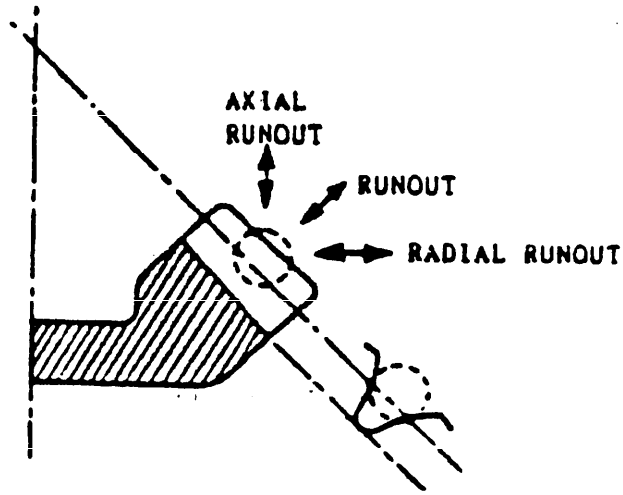


FIGURE 29. Single probe runout check.

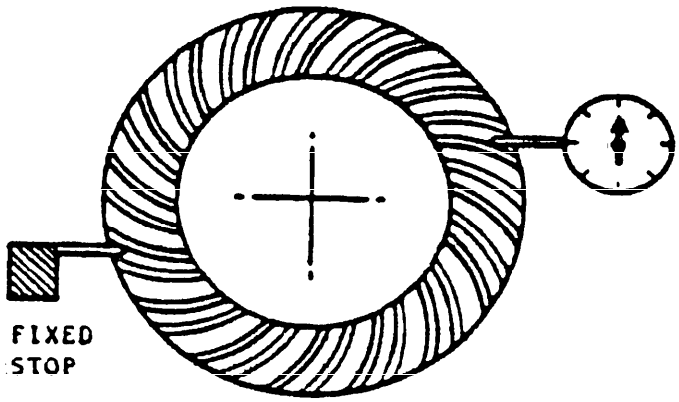


FIGURE 30. Double probe runout check.

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5.4 Assembly of bevel gears.

5.4.1 Purpose. The purpose of this section is to define procedures to be followed when assembling bevel gears into their final mounting positions, and is applicable to bevel gears of the types specified in table VII.

TABLE VII. Bevel gear type.

Item	Bevel		Hypoid	
Gear Type:	Straight	Coniflex	Spiral	Zerol

5.4.2 GENERAL REQUIREMENT

5.4.2.1 Development procedure. The development procedure for all bevel gears includes the determination of desired tooth bearing with both members at their respective reference mounting distances. Each bevel gear drawing specifies the reference mounting distance for that particular gear. This distance is a linear dimension from the axial locating surface (thrust face) of the one gear to the axis of its mate. It is to this reference mounting distance that all bevel gears shall be initially assembled. Assembly should avoid confusing this dimension with various operating distances (normal, minimum and maximum) which will also appear on some bevel gear drawings.

5.4.3 Assembly. The basic assembly procedure shall be as follows:

a. Assemble both members of a gear set at their respective reference mounting distance and within specified backlash limits.

b. Check the tooth bearing with both members mounted as specified in 5.4.2.1.

c. Make any necessary adjustments in original positioning in order to provide desired tooth bearing and specified backlash.

5.4.3.1 Tooth bearing diagram. The bevel gears to be assembled will be furnished with a diagram of the desired tooth bearing pattern as established in the gear development procedure (see 5.4.2.1). This bearing pattern may appear to differ from what is normally considered desireable under full load. This is due to allowance for the deflections of the teeth and mountings under service conditions.

5.4.3.2 Provision for use of shims. The mountings for all gears covered by this standard shall include provisions for shims. In the original design of the gear, the reference mounting distance is fixed on the basis of using some shim stock over the complete range of tolerance build-up in the final mounting.

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5.4.3.3 Determining solid shim thickness. When a solid ground-to-thickness shim is specified, assembly may use the following steps in establishing the final ground thickness of the solid shim.

a. Assembly may use laminated shims behind the pinion and behind the gear to produce the desired tooth bearing and backlash. The laminated shims shall be of the specified inner diameter (ID) and outer diameter (OD) and shall be made up of 0.002 inch laminations.

b. Once the desired tooth bearing and backlash have been attained, one solid shim shall be ground to the total thickness of the laminated shims for each location (behind pinion and behind gear).

c. The solid shims shall be assembled into the designed positions and the laminated shims previously used may then be recycled for future use.

5.4.4 Required procedures and operations.

5.4.4.1 Set pinion at reference mounting distance. Assembly shall proceed by first setting the pinion at its reference mounting distance. This may be done by means of micrometer measurements, gage blocks, special gages, or any other accurate method which the assembler may devise. This setting shall be made while a light axial pressure (finger pressure or equivalent) is exerted against the pinion. This pressure shall be in the direction of increased backlash. Any initial play in the mountings will thereby be taken up. The location shall be made as close to the reference mounting distance as permitted by the shim lamination thickness. For instance, when shims with 0.002 inch laminations are specified, the pinion may be positioned within $\pm$ 0.001 inch of the reference mounting.

5.4.4.2 Set gear at specified backlash. After the pinion is in place, the gear shall be properly positioned. This is done by moving the gear axially until the specified backlash is obtained. The backlash shall be checked while a light axial pressure (finger pressure or equivalent) is exerted against both members in the direction of increased backlash. A minimum of four combinations of mating teeth, approximately equally spaced about the periphery of the gears shall be checked. Each bevel gear detail drawing and assembly drawing shall specify the backlash to which the gears shall be assembled. All backlash values specified on drawings refer to normal backlash at the extreme heel of the tooth. Where practical, this measurement shall be made as shown in figure 31. In those cases where it is not possible to measure the backlash directly as shown in figure 31, it will be necessary to measure the shaft rotation to obtain the transverse backlash. Figure 32 depicts the method of obtaining transverse backlash which is converted to normal backlash by calculation. This is accomplished by the following formula. Backlash = indicator reading x cosine of the pressure angle x cosine of the spiral angle. To measure the shaft rotation a fixture

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errors is not greater than the limits shown on the gear

(flag) shall be attached to the shaft of the gear to be measured and a dial indicator or equivalent is positioned so that the probe point is located at a point on the fixture at a distance equal to the pitch radius of the gears. The measurement obtained (transverse backlash) is converted to normal backlash by use of the formula specified on figure 32. A minimum of four checks equally spaced about the periphery of the shaft shall be made.

5.4.4.3 Reverse assembly sequence. In some cases because of accessability problems, it may be impossible to follow the above sequence of assembly. It may prove necessary to reverse the procedure in which case the sequence will be.

a. Position the gear member at its reference mounting distance.

b. After the gear member is in place, position the pinion so that the specified backlash is obtained.

c. It is emphasized that this is not the preferred sequence and should be used only where absolutely necessary.

5.4.4.4 Tooth bearing check. With both gear and pinion fastened in place, the tooth bearing shall be checked. This is done by painting the gears with a non-abrasive marking compound and rotating them in mesh with one another under light brake load. The resulting tooth bearing shall closely duplicate the pattern established in the gear development procedure. If the tooth bearing pattern or the backlash does not meet the specified requirements, adjustments shall be made as specified in 5.4.4.5 through 5.4.4.7.

5.4.4.5 Adjustments for tooth bearing and backlash. If the actual tooth bearing obtained is unacceptable, adjustments may be made in the position at which the pinion was first assembled. Should the backlash be unacceptable after a suitable pattern has been thereby obtained, it may be possible to correct this error by changing the axial position of the gear. In the case of 2:1 ratios and higher, the gear member may be moved axially for backlash adjustment with very little change in the tooth bearing. However, in the case of miters and low ratios, movement of the gear member alone will affect tooth bearing. Compensation for this may be made by axial adjustment of the pinion in the direction opposite to that of the gear, the relative movement being in proportion to the numbers of teeth. The types of bearing deviations for which axial adjustments may be made are given together with the necessary corrective procedures:

a. Bearing too high on pinion and too low on gear. This is due to the pinion being too near the axis of the gear member. Correction is made by increasing the pinion mounting distance.

b. Bearing too low on pinion and too high on gear. This is due to the pinion being too far from the axis of the gear member. Correction is made by decreasing the pinion mounting distance.

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5.4.4.6 Mounting errors. In addition, there are several common tooth bearing deviations which are caused by errors in the mountings. These types of deviations, must be corrected by elimination of the problems which cause mounting misalignments.

a. A cross bearing mounting error is one in which the contact is too close to one end of the tooth on one side and too close to the opposite end on the other side. This results from the shafts being non-coplanar.

b. Tooth bearing too far toward same end of teeth on both sides. This results from inaccuracy in the shaft angle. If the bearing area is too close to the toe (small end), the shaft angle is too large. If the bearing area is too close to the heel (large end), the shaft angle is too small. Figure 33 and figure 34 illustrate the aforementioned types of bearing errors along with the remedies. Although these figures use a spiral tooth form for illustrating purposes, the information given and principles involved also apply directly to zerol bevels and coniflex-type straight bevels. The particular bearings illustrated in figure 33 and figure 34 as the correct bearing are given to serve only as fixed references to illustrate direction of deviation. Every spiral, zerol, coniflex and straight bevel gear will have its own correct bearing which is fixed by its discrete service conditions.

5.4.4.7 Tooth bearing for straight bevel gears. The tooth bearing pattern for a pair of correctly made straight bevel gears is illustrated on figure 35, and is applicable to all straight bevel gears regardless of service conditions. Figure 35 also illustrates a variety of bearing errors and the corrective action necessary to obtain the correct tooth bearing.

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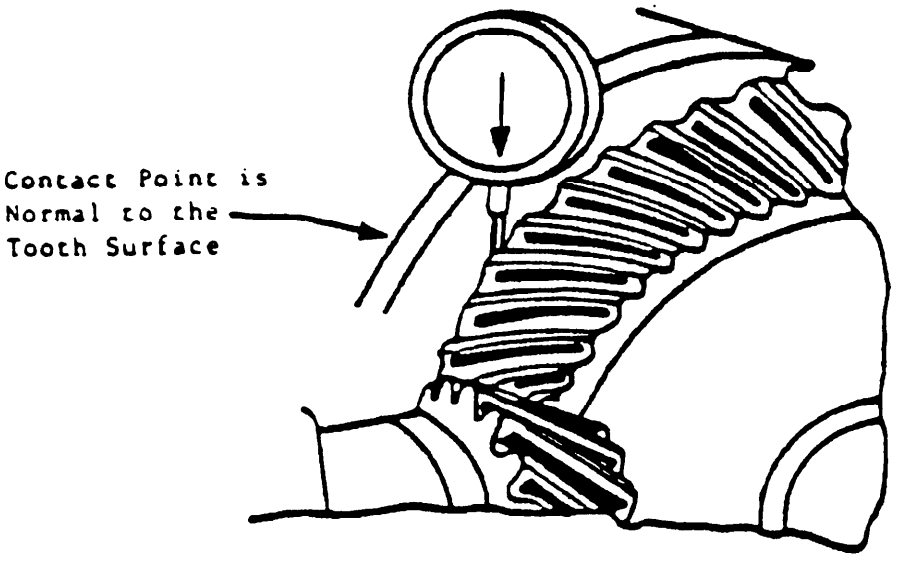
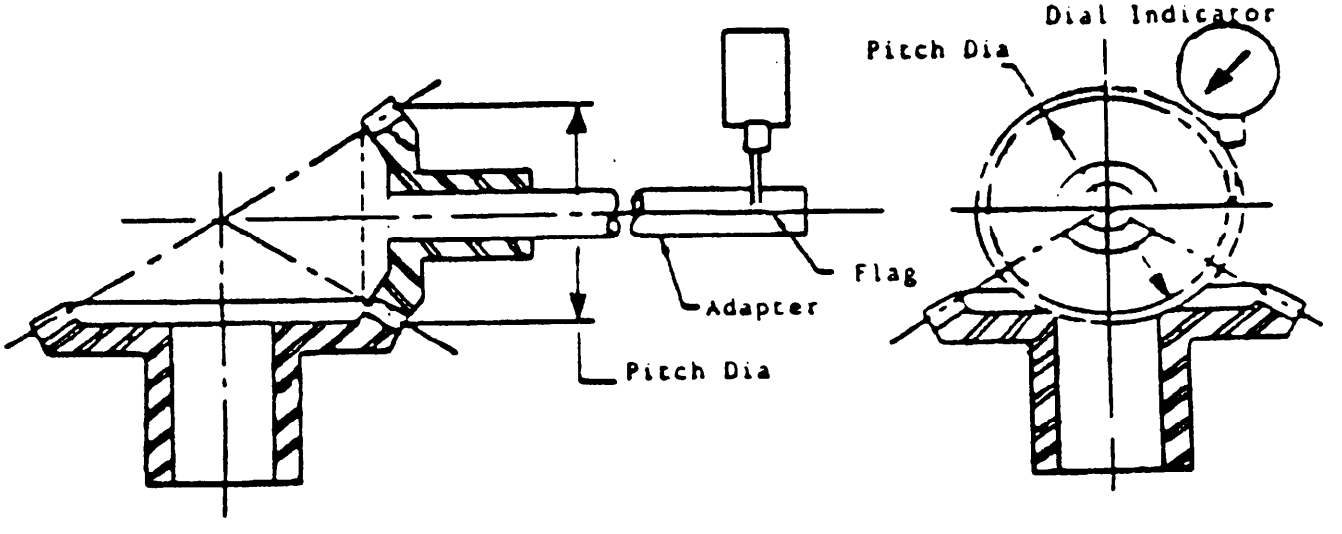


FIGURE 31. Direct method for backlash measurement.



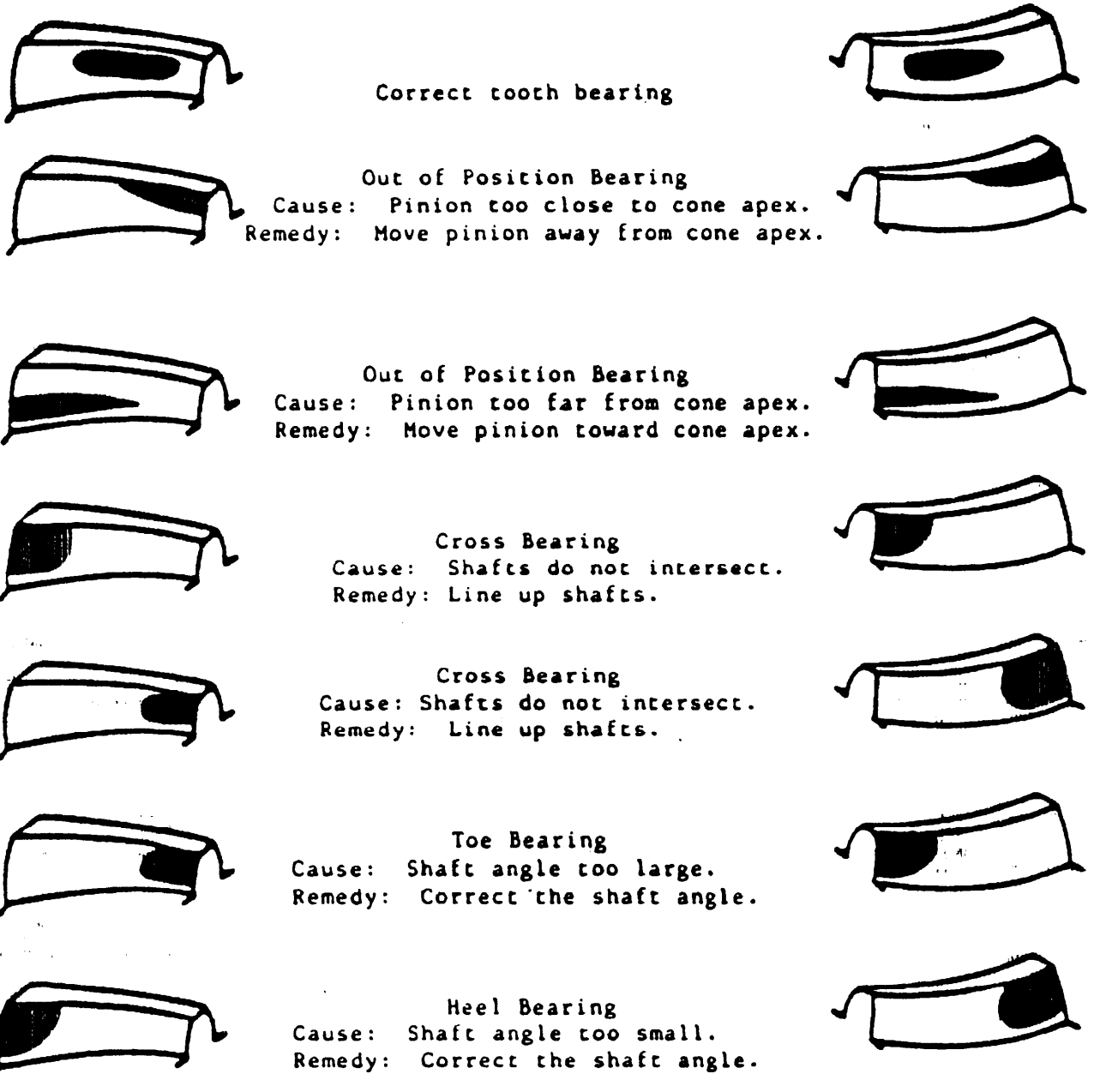
Backlash=Indicator Rdg. x cos. press angle x cos. spiral angle

FIGURE 32. Use of shaft rotation for backlash measurement.

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Concave Side

Convex Side



Correct tooth bearing

Out of Position Bearing
Cause: Pinion too close to cone apex.
Remedy: Move pinion away from cone apex.

Out of Position Bearing
Cause: Pinion too far from cone apex.
Remedy: Move pinion toward cone apex.

Cross Bearing
Cause: Shafts do not intersect.
Remedy: Line up shafts.

Cross Bearing
Cause: Shafts do not intersect.
Remedy: Line up shafts.

Toe Bearing
Cause: Shaft angle too large.
Remedy: Correct the shaft angle.

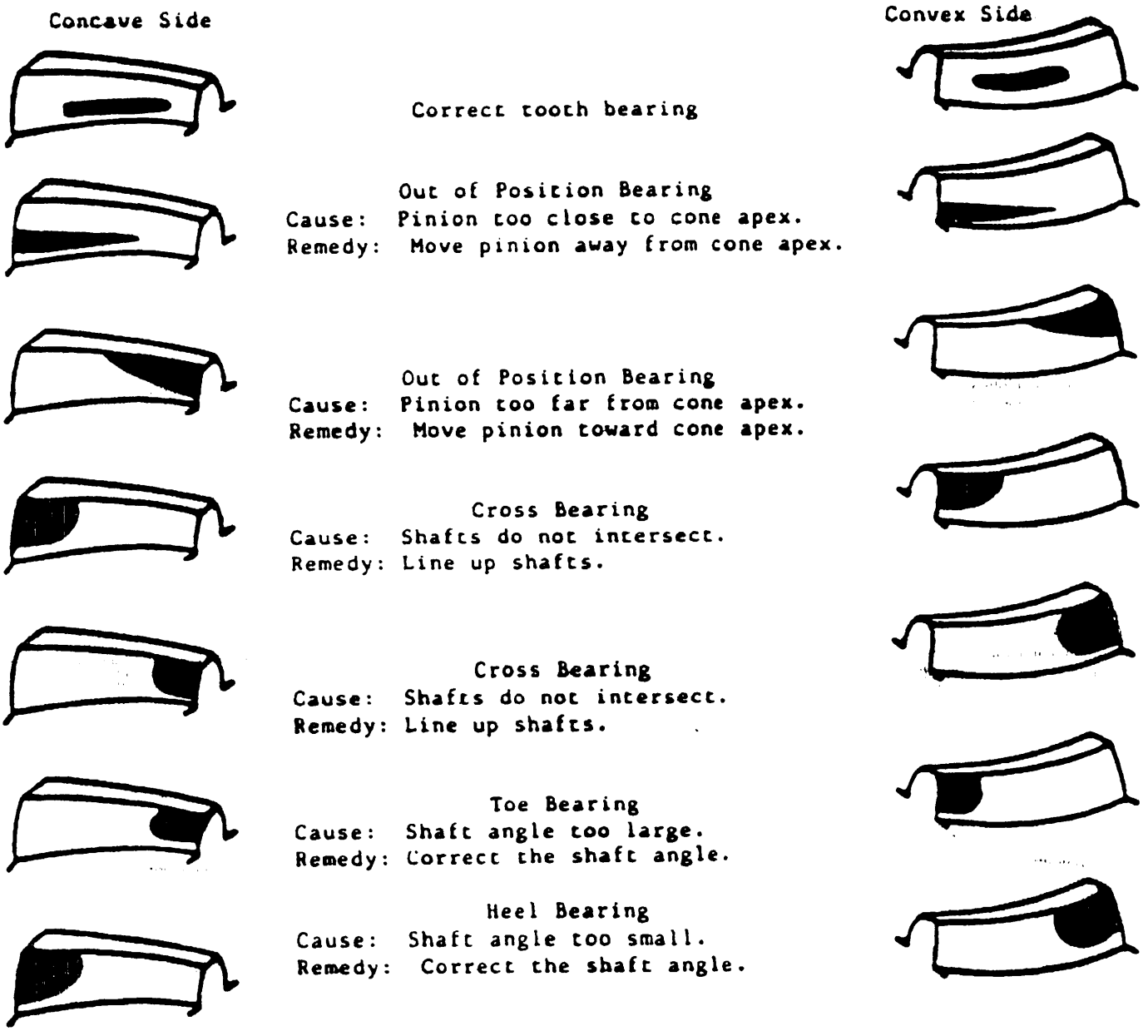
Heel Bearing
Cause: Shaft angle too small.
Remedy: Correct the shaft angle.

Note:

These sketches show the possible tooth bearings on the pinion member of a pair of correctly-made spiral, zerol or coniflex straight bevel gears. The interpretation of these sketches is clarified in 4.3. No use shall be made of this information without having first read 4.3.

FIGURE 33. Tooth bearing on pinions, zerol, spiral and Coniflex.

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Note:

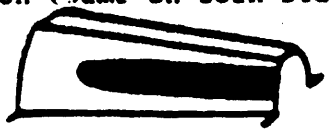
These sketches show the possible tooth bearings on the gear member of a pair of correctly-made sprial, zerol, or coniflex straight bevel gears. The interpretation of these sketches is clarified in 4.3. No use shall be made of this information without having first read 4.3.

FIGURE 34. Tooth bearings on pinions, zerol, spiral and Coniflex.

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(a) Pinion (same on both sides)

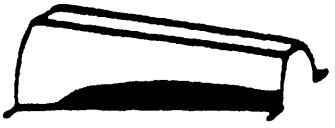


(b) Gear (same on both sides)

Correct Tooth Bearing

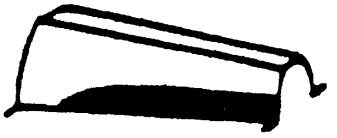


(a) Pinion (same on both sides)



(b) Gear (same on both sides)

Out of Position Bearing
Cause: Pinion to close to cone apex.
Remedy: Move pinion away from cone apex.

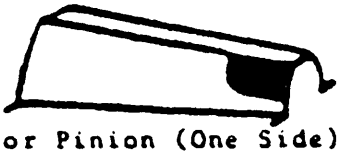


(a) Pinion (same on both sides)



(b) Gear (same on both sides)

Out of Position Bearing
Cause: Pinion too far from cone apex.
Remedy: Move pinion toward cone apex.



(a) Gear or Pinion (One Side)



(b) Gear or Pinion (Opposite Side)

Cross Bearing
Cause: Shafts do not intersect.
Remedy: Line up shafts



(a) Gear or Pinion (Both Sides)



(b) Gear or Pinion (Both Sides)

Toe or Heel Bearing
Cause:
Toe Bearing: Shaft angle too large.
Heel Bearing: Shaft angle too small
Remedy: Correct the shaft angle.

Note:

These sketches show the possible tooth bearings for a pair of correctly-made straight bevel gears which are not of the coniflex type. The correct tooth bearing shown above applies to all straight bevel gears of this type regardless of differing service conditions.

FIGURE 35. Tooth bearings on straight bevel gears and pinions.

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6. NOTES

6.1 Supersession data. This document supersedes MIL-G-62454, dated 13 July 1983; AVCO Lycoming Division specifications E2152A, dated 26 April 1982; P6805B, dated 27 September 1979; P6920B, dated 27 September 1979; and P7800C, dated 27 September 1979.

6.2 Subject term (key word) listing.

Bevel gears
Gears
Spur gears

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 3020-A006)

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" 2.1 Lot formation. "The term "LOT" as used throughout

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